

Surprise Minimization Experimental Results

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Introducing the Deliverables

SEED_INIT	1	N	30
T	100	DT	0.01
N_SECTORS	4	SECTOR_ANGLE	60.0
DIST_THR	5.0	Z_H	0.01
Z_HPRIME	0.01	Z_ACTION	0.01
ALPHA	0.05	ETA	1.0
PI_Z_SPATIAL	1.0	PI_W_SPATIAL	1.0
S_Z	1.0	S_W	1.0
INFER_LR	0.1	ACTION_LR	0.1
NSTEPS_INFER	1	NSTEPS_ACTION	1
NORMALIZE_V	True	SEED_ERROR	None

The following content was generated using different branches of the Jax_Backend code from collective_motion_actinf. The recommended branch for recreating these visualization is:

“experiment_random_neighbor_per_sector”.

Any titles containing “Avg Neighbor” were run with RANDOM_NEIGHBOR = False

Any titles containing “One Neighbor” were run with RANDOM_NEIGHBOR = True

The initial parameter set up can be seen in the table on the left.

***Note SEED_ERROR was given a numerical value for all runs.*

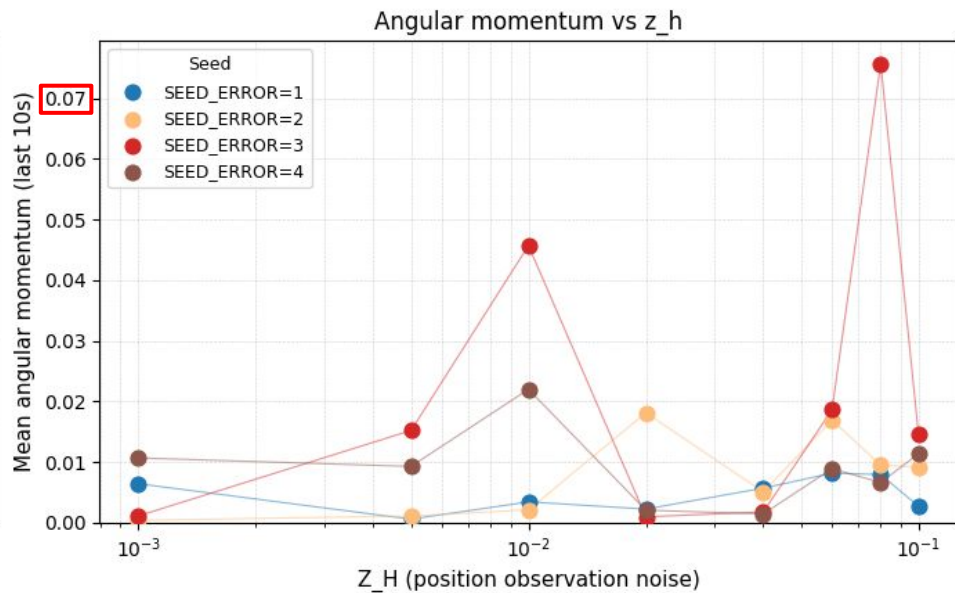
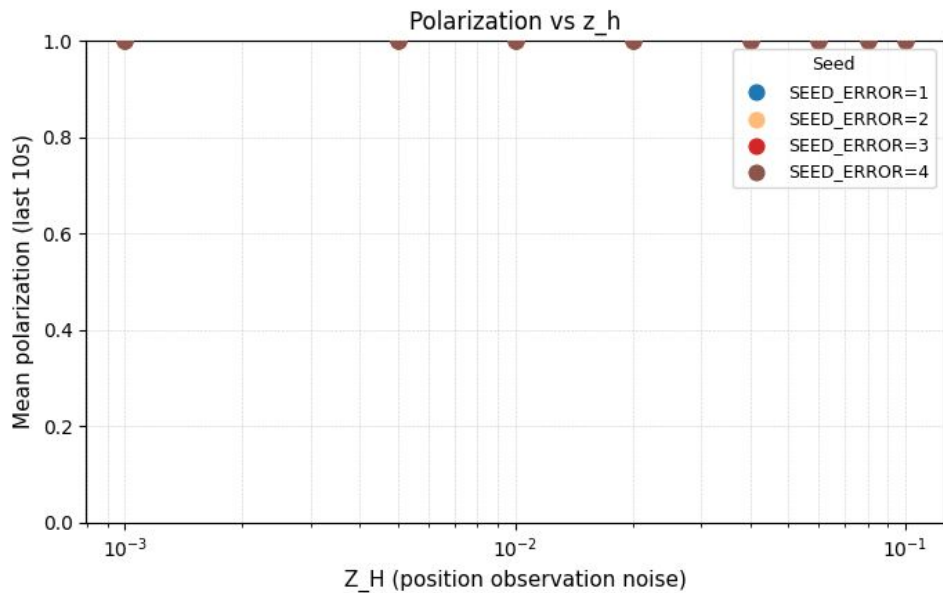
Z_H , $Z_{H'}$ & Z_{Action} across Error Seeds (Fixed Initial) – Avg Neighbor

SEED_INIT	1	N	30
T	100	DT	0.01
N_SECTORS	4	SECTOR_ANGLE	60.0
DIST_THR	5.0	Z_H	0.0
Z_{HPRIME}	0.0	Z_{ACTION}	0.0
ALPHA	0.05	ETA	1.0
PI_Z_SPATIAL	1.0	PI_W_SPATIAL	1.0
S_Z	1.0	S_W	1.0
INFER_LR	0.1	ACTION_LR	0.1
NSTEPS_INFER	1	NSTEPS_ACTION	1
NORMALIZE_V	True	SEED_ERROR	Seed

The left table contains the parameter set up for the following visualizations. When the Z noise were not swept, they were set to the value 0.

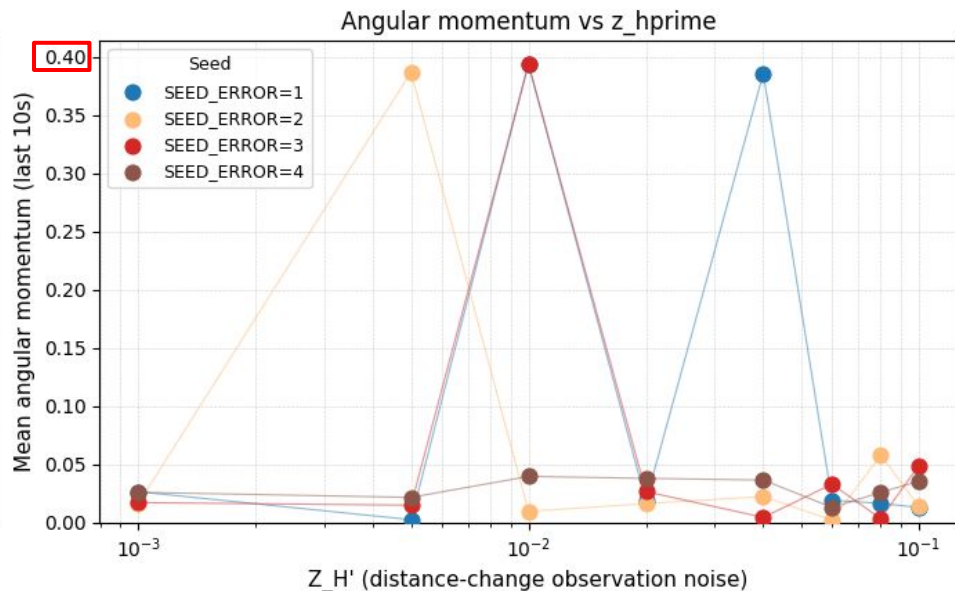
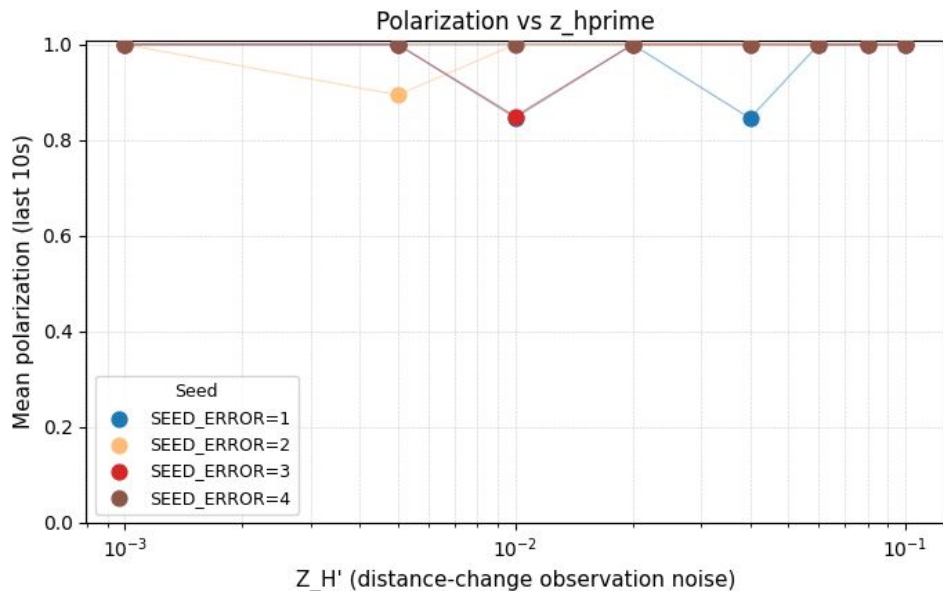
Z_H across Error Seeds (Fixed Initial) – Avg Neighbor

Effect of Z_H (position observation noise) on collective behavior
($N=30$, $T=100$, $SEED_INIT=1$, other noise params set to 0.0)



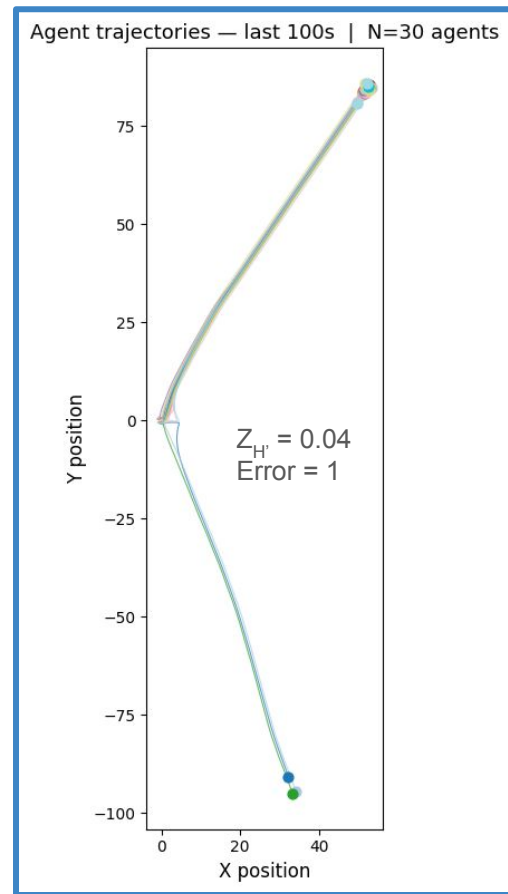
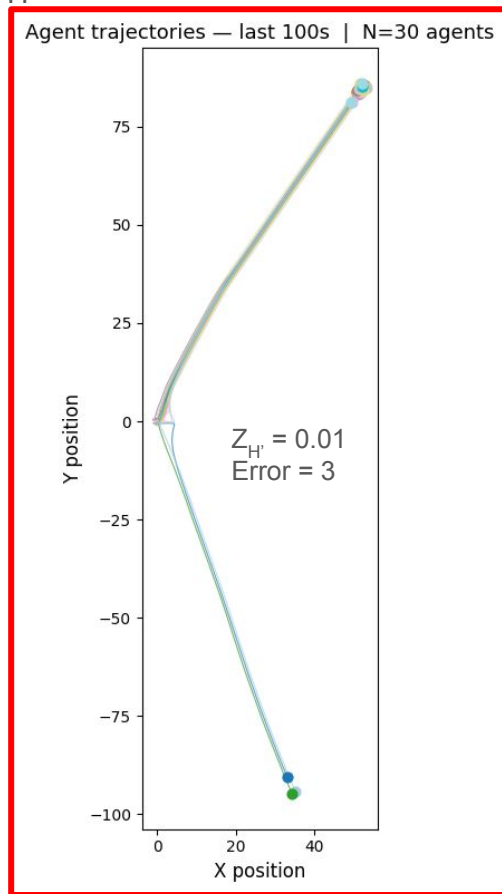
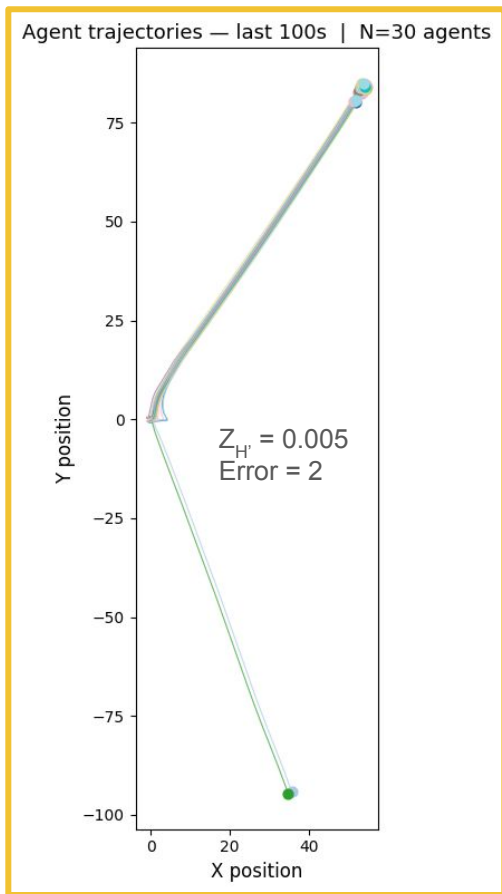
Z_H , across Error Seeds (Fixed Initial) – Avg Neighbor

Effect of Z_H ' (distance-change observation noise) on collective behavior
($N=30$, $T=100$, $SEED_INIT=1$, other noise params set to 0.0)



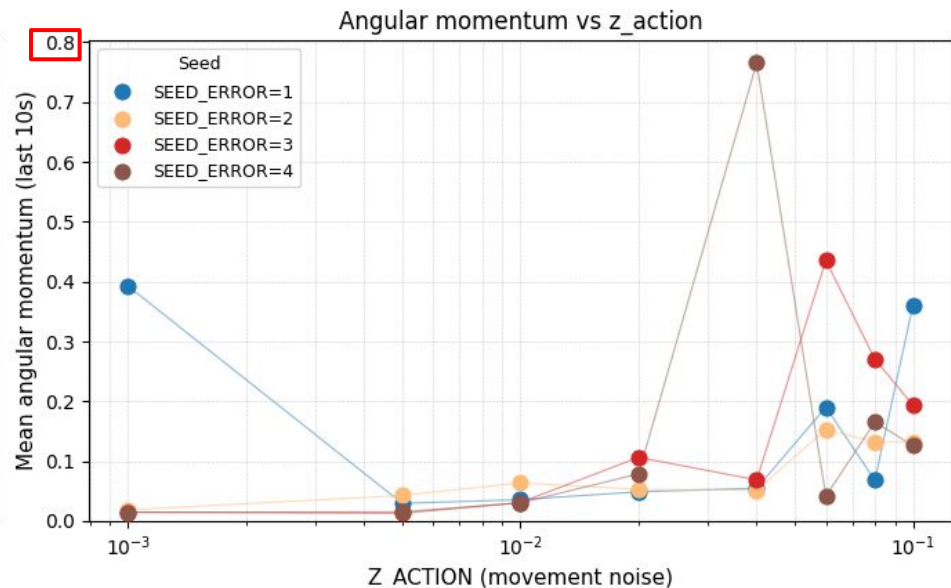
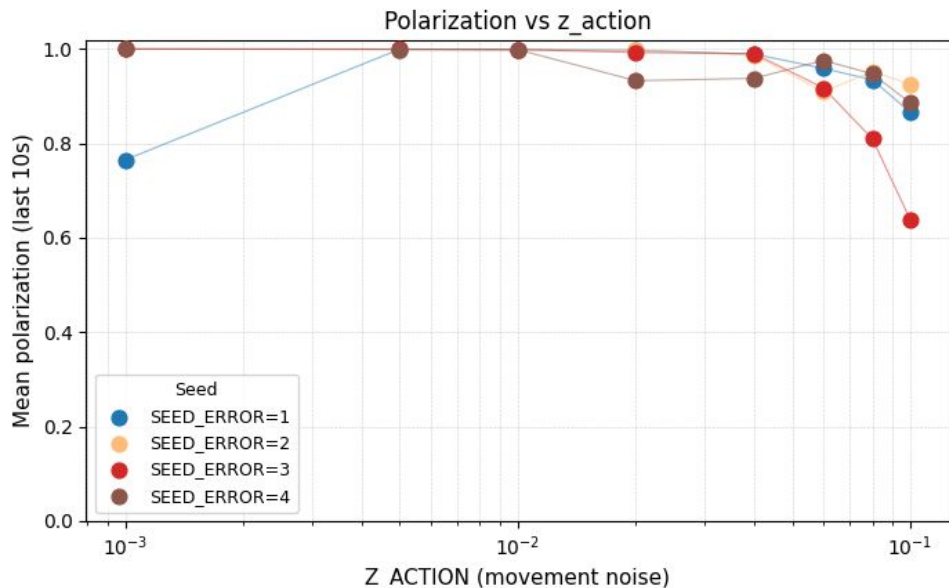
$Z_{H'}$, across Error Seeds (Fixed Initial) – Avg Neighbor

Specific visualizations of certain $Z_{H'}$ and Error Seed combinations:



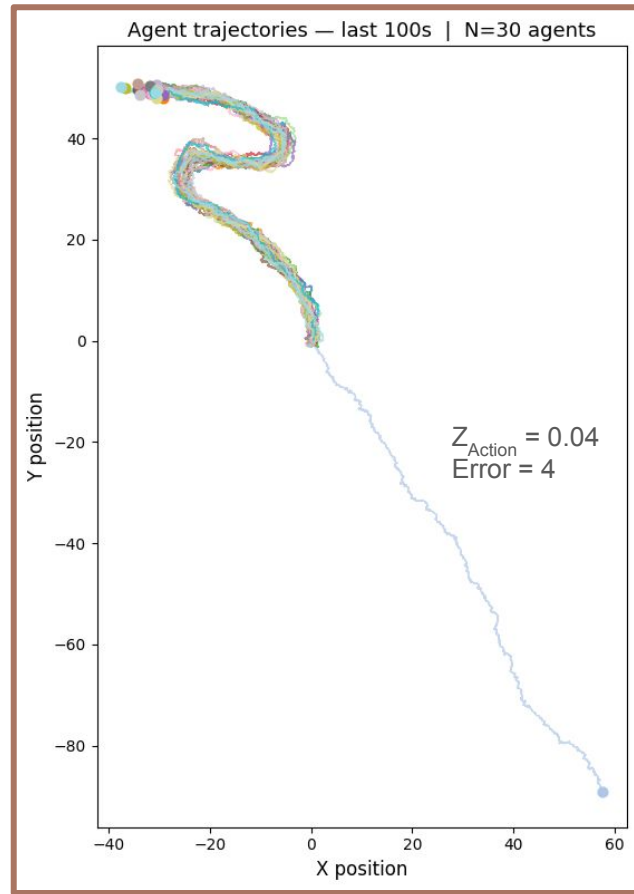
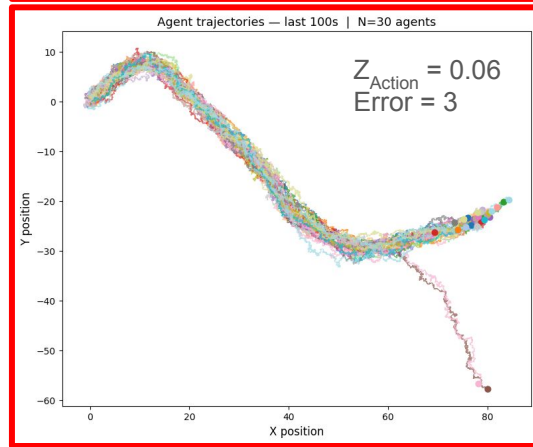
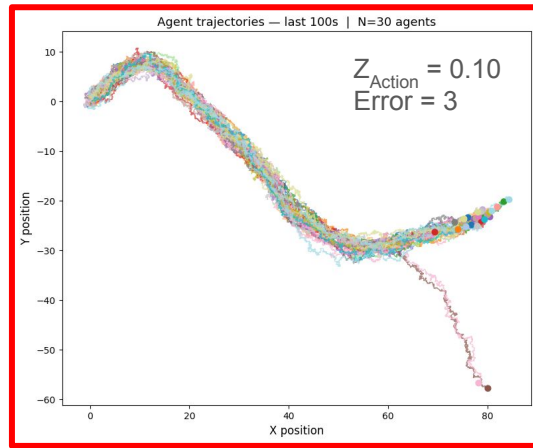
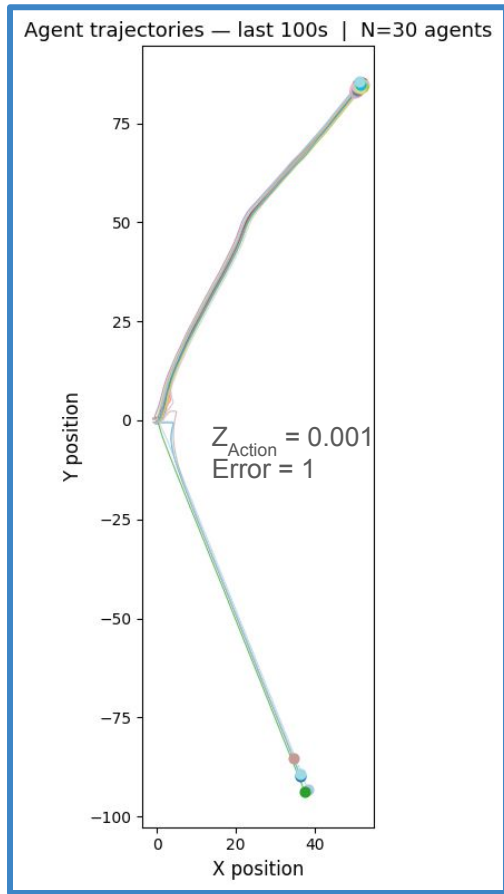
Z_{Action} across Error Seeds (Fixed Initial) – Avg Neighbor

Effect of Z_{ACTION} (movement noise) on collective behavior
($N=30$, $T=100$, $\text{SEED_INIT}=1$, other noise params set to 0.0)



Z_{Action} across Error Seeds (Fixed Initial) – Avg Neighbor

Specific visualizations of certain Z_{Action} and Error Seed combinations:



Z_H , $Z_{H'}$ & Z_{Action} across Error Seeds (Fixed Initial) – One Neighbor

SEED_INIT	1	N	30
T	100	DT	0.01
N_SECTORS	4	SECTOR_ANGLE	60.0
DIST_THR	5.0	Z_H	0.0
Z_{HPRIME}	0.0	Z_{ACTION}	0.0
ALPHA	0.05	ETA	1.0
PI_Z_SPATIAL	1.0	PI_W_SPATIAL	1.0
S_Z	1.0	S_W	1.0
INFER_LR	0.1	ACTION_LR	0.1
NSTEPS_INFER	1	NSTEPS_ACTION	1
NORMALIZE_V	True	SEED_ERROR	Seed

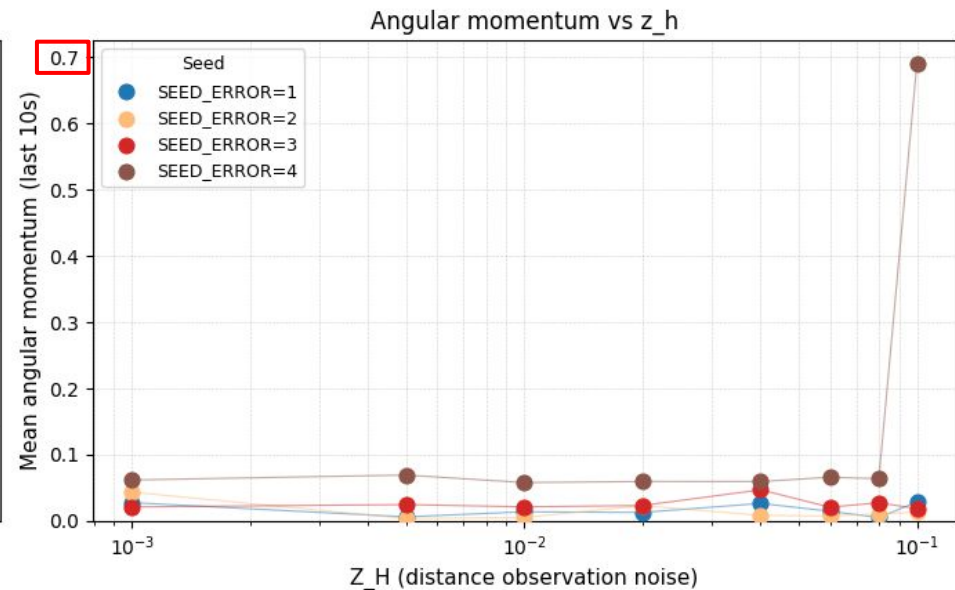
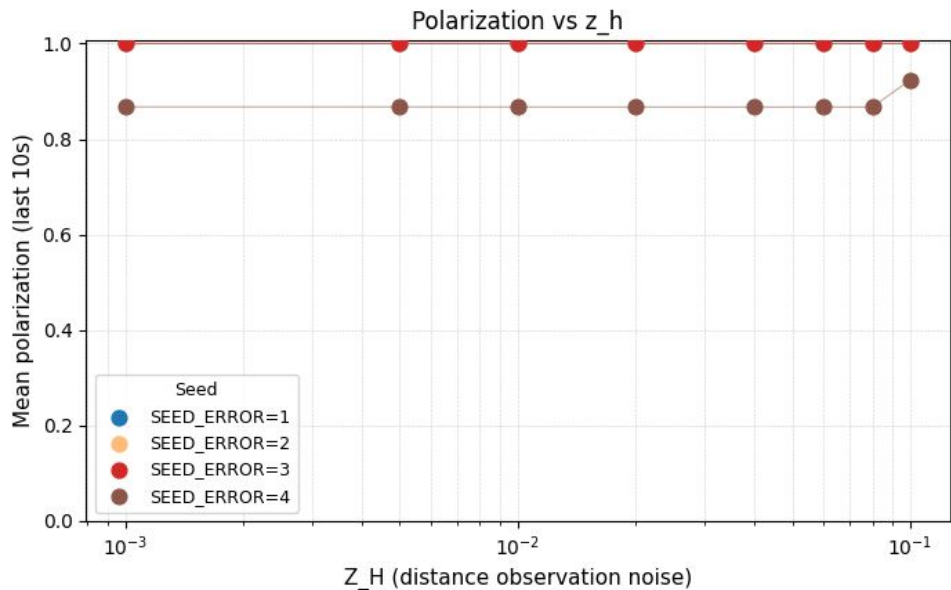
The left table contains the parameter set up for the following visualizations. When the Z noise values were not swept, they were set to the value 0.

Note for these runs:

RANDOM_NEIGHBOR = True

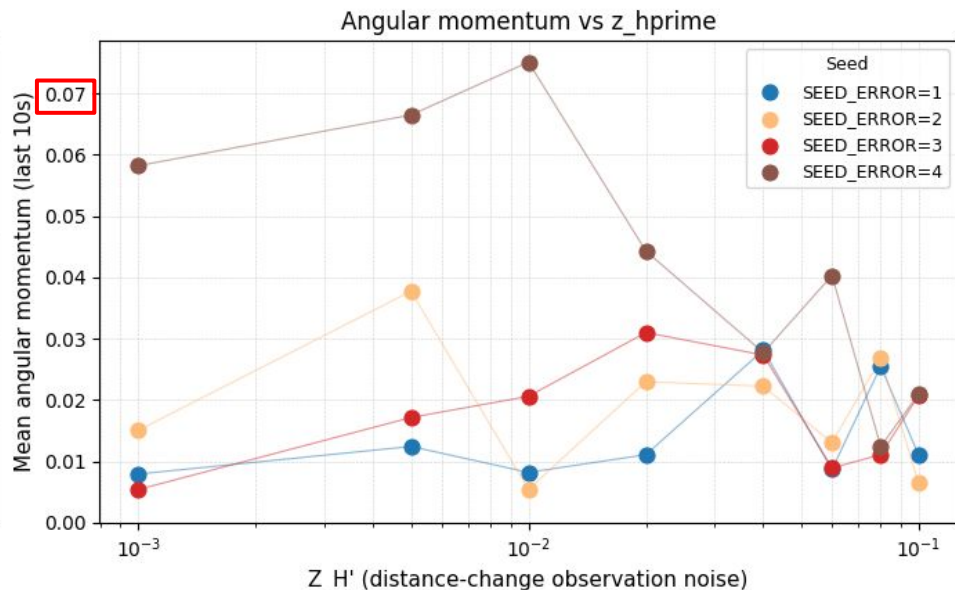
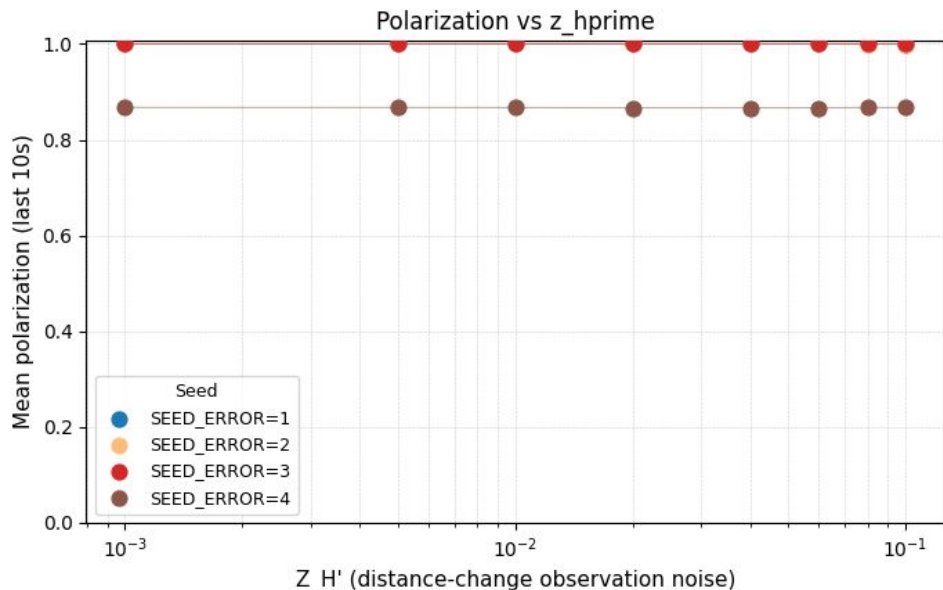
Z_H across Error Seeds (Fixed Initial) – One Neighbor

Effect of Z_H (distance observation noise) on collective behavior
($N=30$, $T=100$, $SEED_INIT=1$, other noise params set to 0.0, random neighbor per sector)



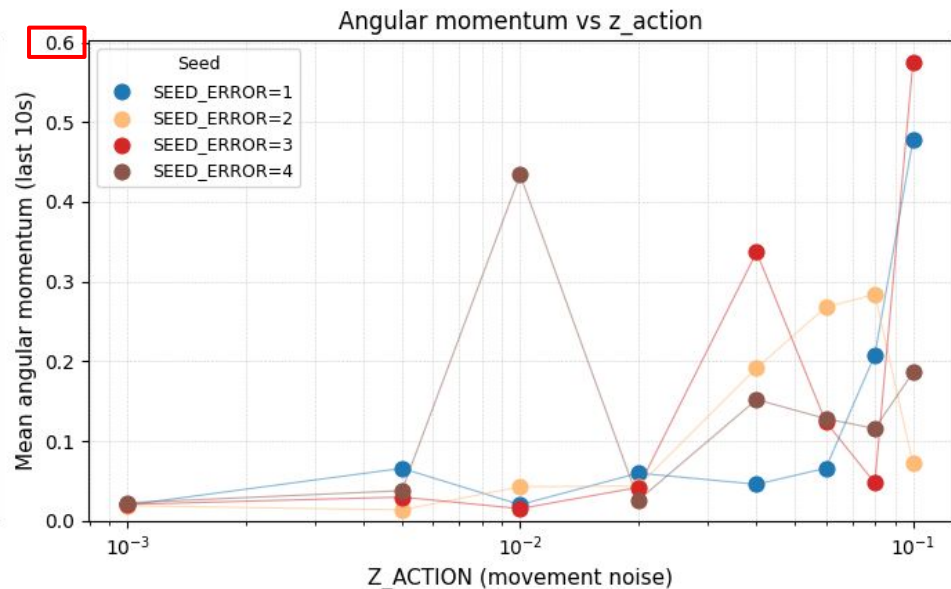
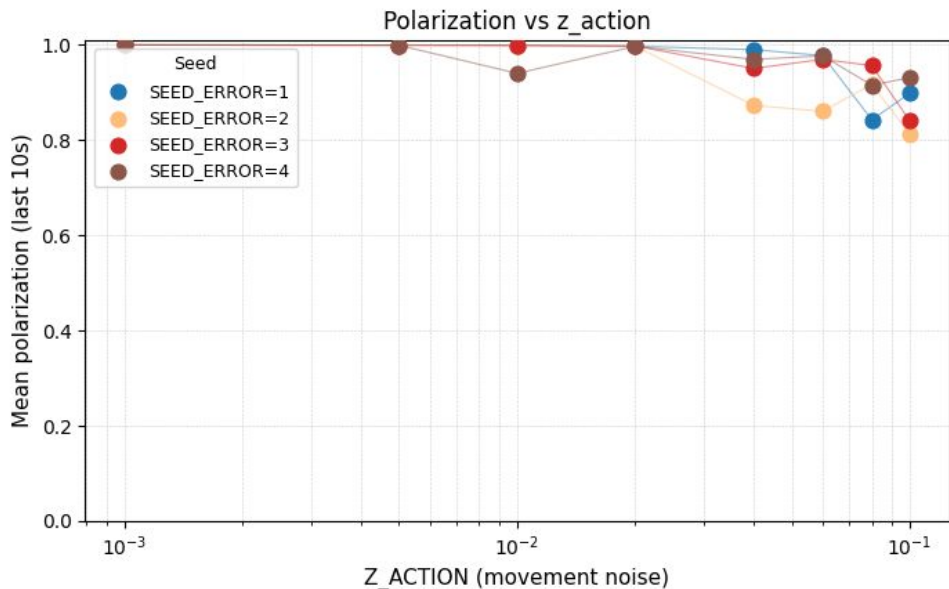
Z_H , across Error Seeds (Fixed Initial) – One Neighbor

Effect of Z_H ' (distance-change observation noise) on collective behavior
($N=30$, $T=100$, $SEED_INIT=1$, other noise params set to 0.0, random neighbor per sector)



Z_{Action} across Error Seeds (Fixed Initial) – One Neighbor

Effect of Z_{ACTION} (movement noise) on collective behavior
($N=30$, $T=100$, $\text{SEED_INIT}=1$, other noise params set to 0.0, random neighbor per sector)



Z_H , $Z_{H'}$ & Z_{Action} across Error Seeds (Fixed Initial) – Comparison

SEED_INIT	1	N	30
T	100	DT	0.01
N_SECTORS	4	SECTOR_ANGLE	60.0
DIST_THR	5.0	Z_H	0.0
Z_{HPRIME}	0.0	Z_{ACTION}	0.0
ALPHA	0.05	ETA	1.0
PI_Z_SPATIAL	1.0	PI_W_SPATIAL	1.0
S_Z	1.0	S_W	1.0
INFER_LR	0.1	ACTION_LR	0.1
NSTEPS_INFER	1	NSTEPS_ACTION	1
NORMALIZE_V	True	SEED_ERROR	Seed

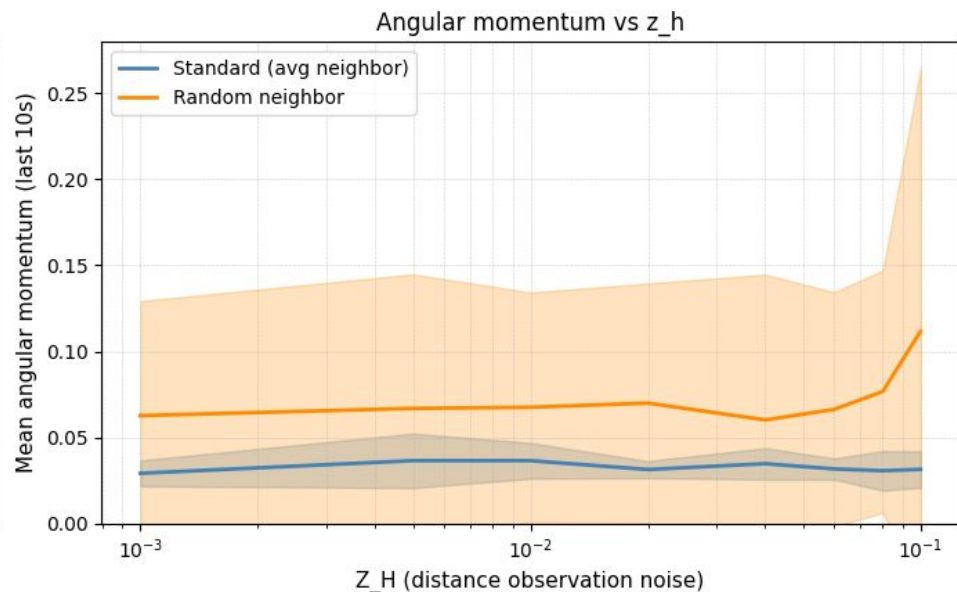
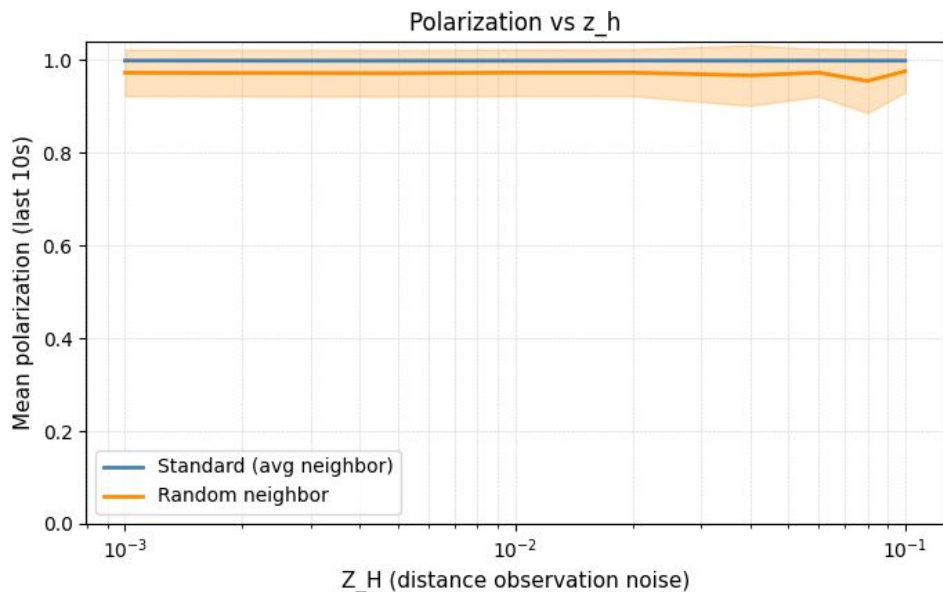
The left table contains the parameter set up for the following visualizations. When the Z noise values were not swept, they were set to the value 0.

The following graphs show the average polarization values across the four error [noise realization] seeds (from the previous graphs) and compares the output with and without the random neighbor sampling style.

The shaded portion represents the standard deviation value.

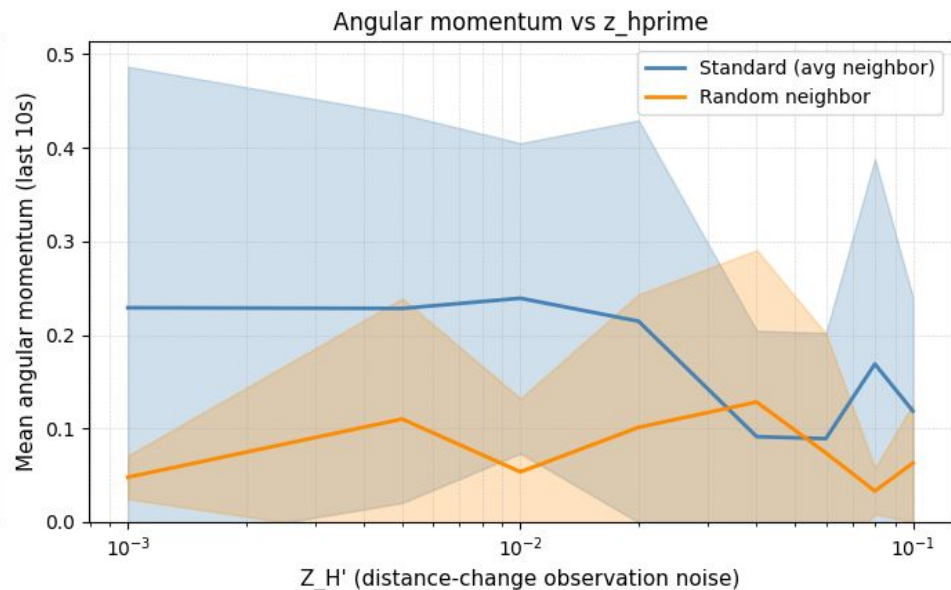
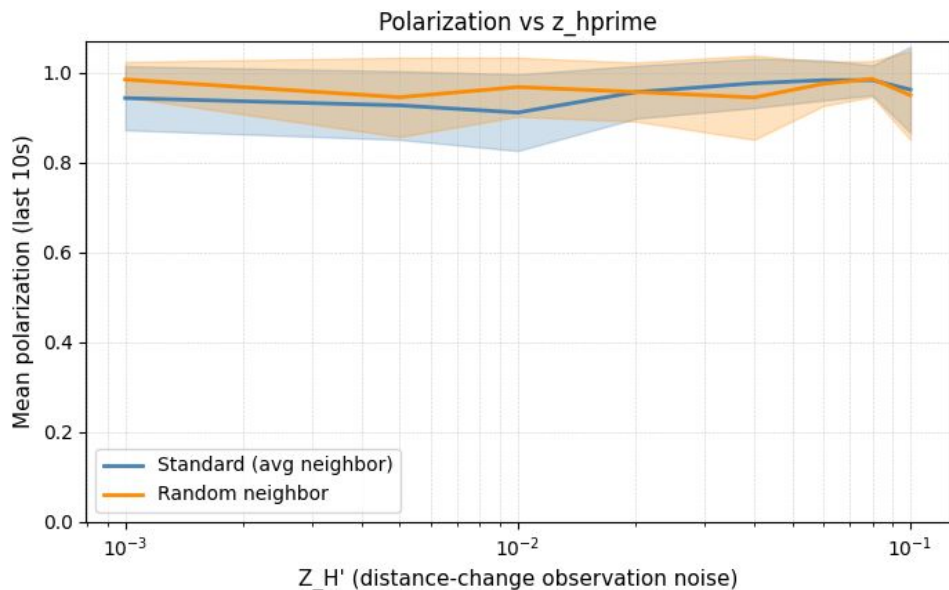
Z_H across Error Seeds (Fixed Initial) – Comparison

Standard vs random neighbor — effect of Z_H (distance observation noise)
($N=30$, $T=10$, $SEED_INIT=1$, other noise=0.0, mean $\pm$ std across 10 error seeds)



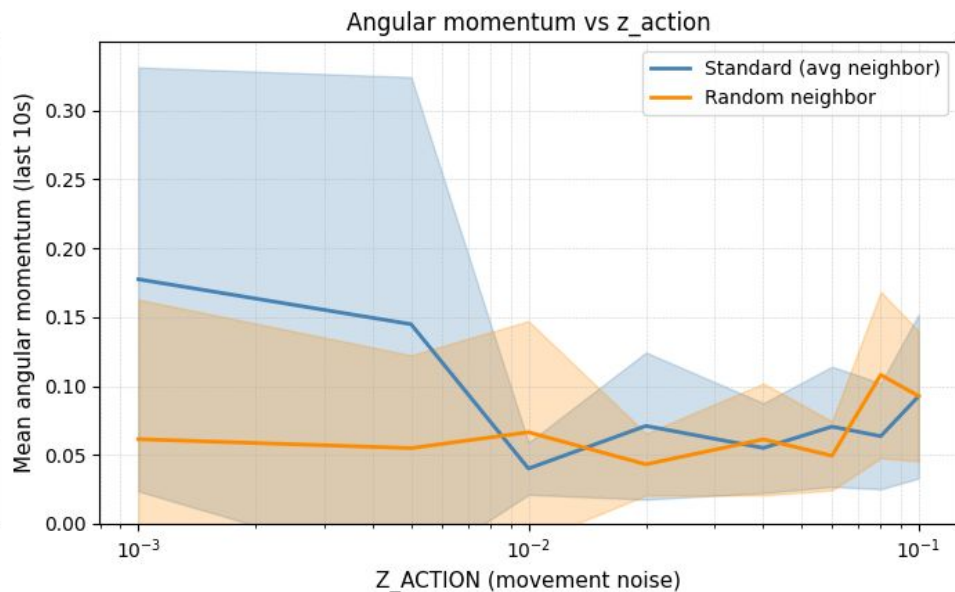
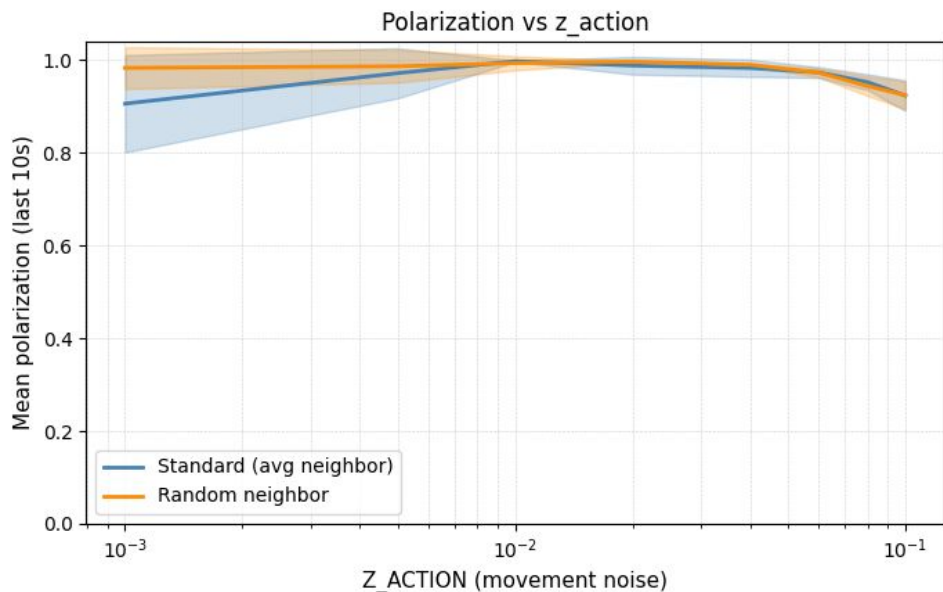
Z_H , across Error Seeds (Fixed Initial) – Comparison

Standard vs random neighbor — effect of Z_H ' (distance-change observation noise)
($N=30$, $T=10$, $SEED_INIT=1$, other noise=0.0, mean $\pm$ std across 10 error seeds)



Z_{Action} across Error Seeds (Fixed Initial) – Comparison

Standard vs random neighbor — effect of Z_{ACTION} (movement noise)
($N=30$, $T=10$, $\text{SEED_INIT}=1$, other noise=0.0, mean $\pm$ std across 10 error seeds)



Π_Z, Π_W, S_Z, S_W across Error Seeds (Fixed Initial) – Avg Neighbor

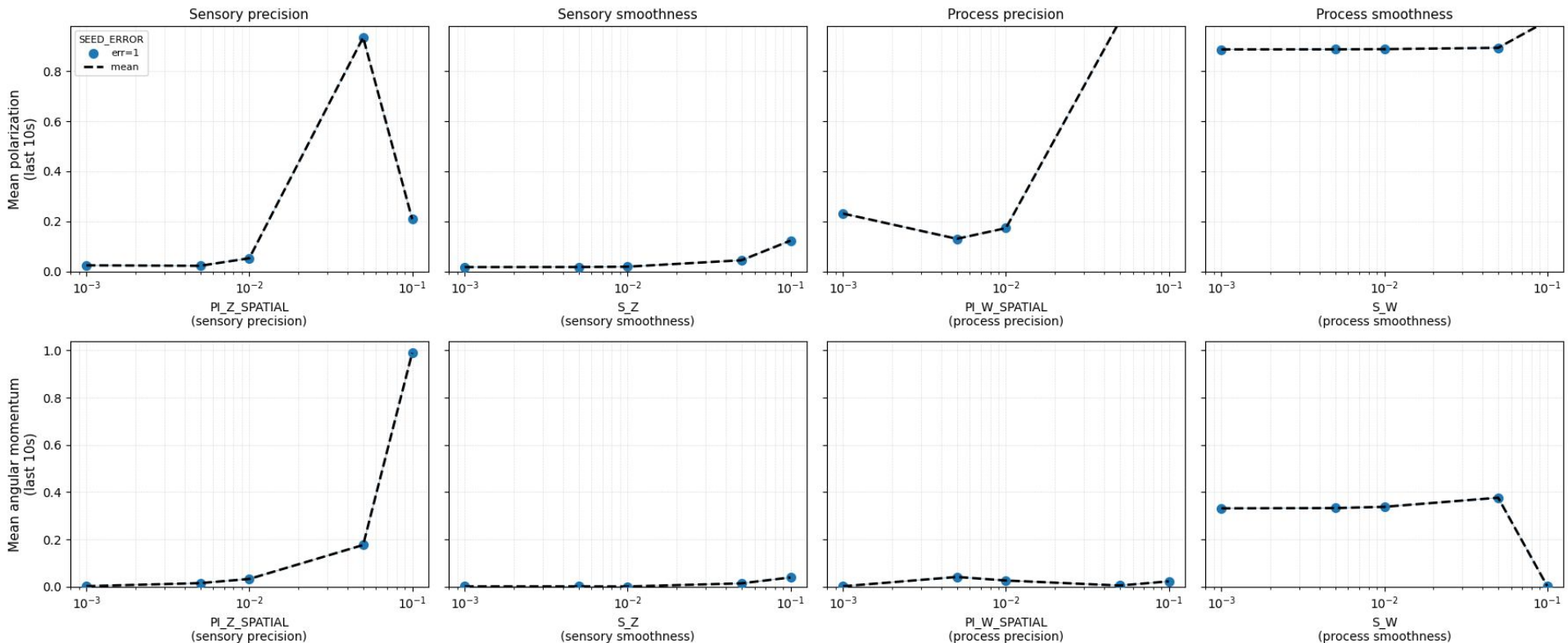
SEED_INIT	1	N	30
T	100	DT	0.01
N_SECTORS	4	SECTOR_ANGLE	60.0
DIST_THR	5.0	Z_H	0.0
Z_HPRIME	0.0	Z_ACTION	0.0
ALPHA	0.05	ETA	1.0
PI_Z_SPATIAL	1.0	PI_W_SPATIAL	1.0
S_Z	1.0	S_W	1.0
INFER_LR	0.1	ACTION_LR	0.1
NSTEPS_INFER	1	NSTEPS_ACTION	1
NORMALIZE_V	True	SEED_ERROR	Seed

The left table contains the parameter set up for the following visualizations. When the highlighted parameters were not swept, they retained the value 1.0 as shown in the table.

These visualizations were not included in the final deliverable but are interesting to consider.

Π_Z, Π_W, S_Z, S_W across Error Seeds (Fixed Initial) – Avg Neighbor

Effect of internal model parameters on collective behavior
($N=30, T=100, \text{SEED_INIT}=1, Z_H=0.0, Z_{HPRIME}=0.0, Z_{\text{ACTION}}=0.0$)



Π_Z, Π_W, S_Z, S_W across Error Seeds (Fixed Initial) – One Neighbor

SEED_INIT	1	N	30
T	100	DT	0.01
N_SECTORS	4	SECTOR_ANGLE	60.0
DIST_THR	5.0	Z_H	0.0
Z_HPRIME	0.0	Z_ACTION	0.0
ALPHA	0.05	ETA	1.0
PI_Z_SPATIAL	1.0	PI_W_SPATIAL	1.0
S_Z	1.0	S_W	1.0
INFER_LR	0.1	ACTION_LR	0.1
NSTEPS_INFER	1	NSTEPS_ACTION	1
NORMALIZE_V	True	SEED_ERROR	Seed

The left table contains the parameter set up for the following visualizations. When the highlighted parameters were not swept, they retained the value 1.0 as shown in the table.

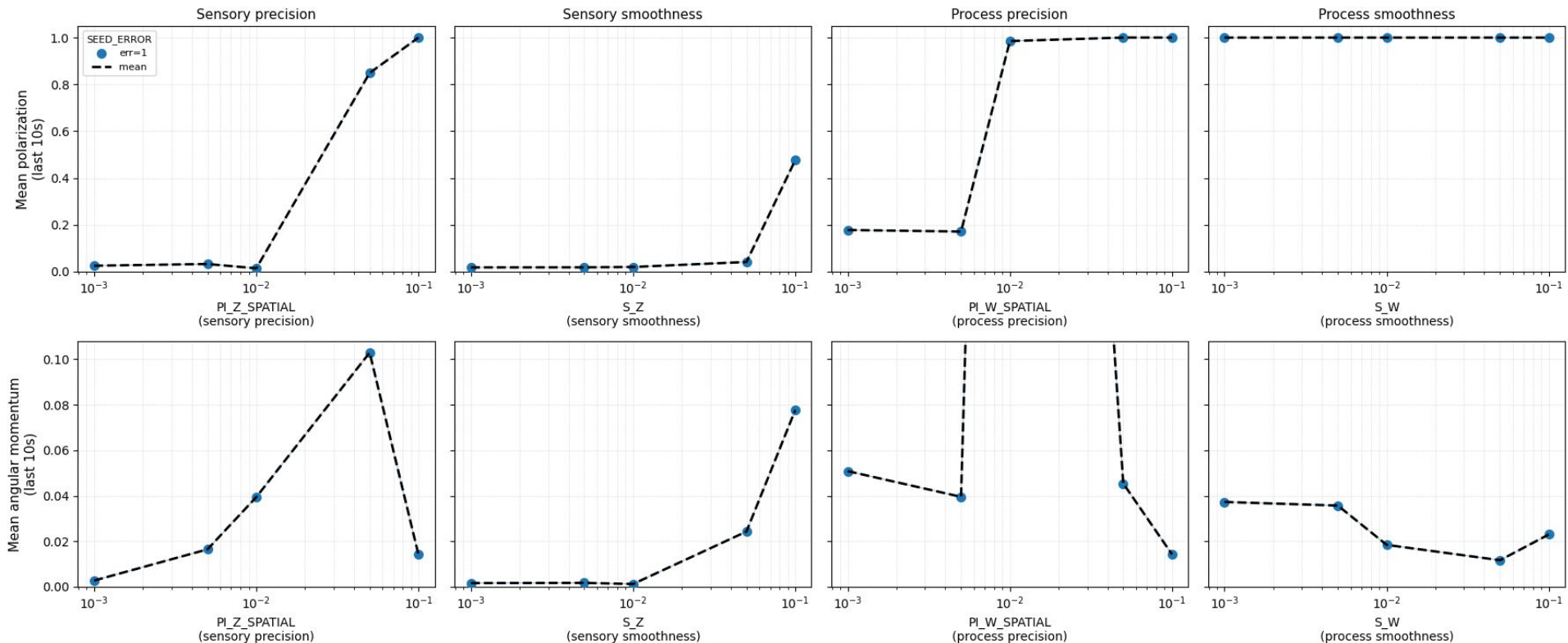
These visualizations were not included in the final deliverable but are interesting to consider.

Note for these runs:

RANDOM_NEIGHBOR = True

Π_Z, Π_W, S_Z, S_W across Error Seeds (Fixed Initial) – One

Effect of internal model parameters on collective behavior
($N=30, T=100, \text{SEED_INIT}=1, Z_H=0.0, Z_{HPRIME}=0.0, Z_{\text{ACTION}}=0.0$)



Z_H , $Z_{H'}$, Z_{Action} – $p(t)$ and $m(t)$ plots across time

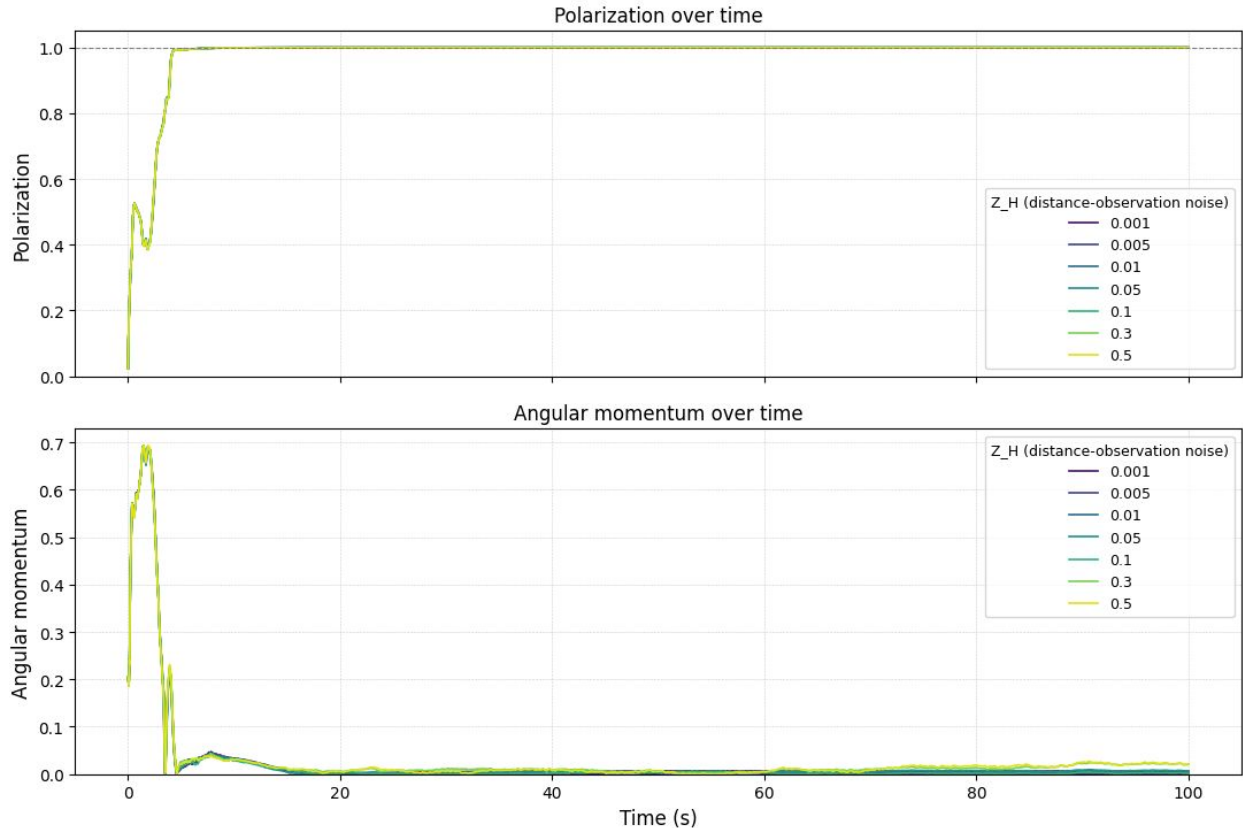
SEED_INIT	1	N	30
T	100	DT	0.01
N_SECTORS	4	SECTOR_ANGLE	60.0
DIST_THR	5.0	Z_H	0.0
Z_{HPRIME}	0.0	Z_{ACTION}	0.0
ALPHA	0.05	ETA	1.0
PI_Z_SPATIAL	1.0	PI_W_SPATIAL	1.0
S_Z	1.0	S_W	1.0
INFER_LR	0.1	ACTION_LR	0.1
NSTEPS_INFER	1	NSTEPS_ACTION	1
NORMALIZE_V	True	SEED_ERROR	1

The left table contains the parameter set up for the following visualizations. When the Z errors were not swept, they were set to the value 0.

Please Note the first three runs lasted 100 seconds and the next six lasted 10 seconds and include both the average and random neighbor versions. The 10 second runs were used in the final presentation for this project.

$Z_H - p(t)$ and $m(t)$ plots across time (100 sec) – Avg Neighbor

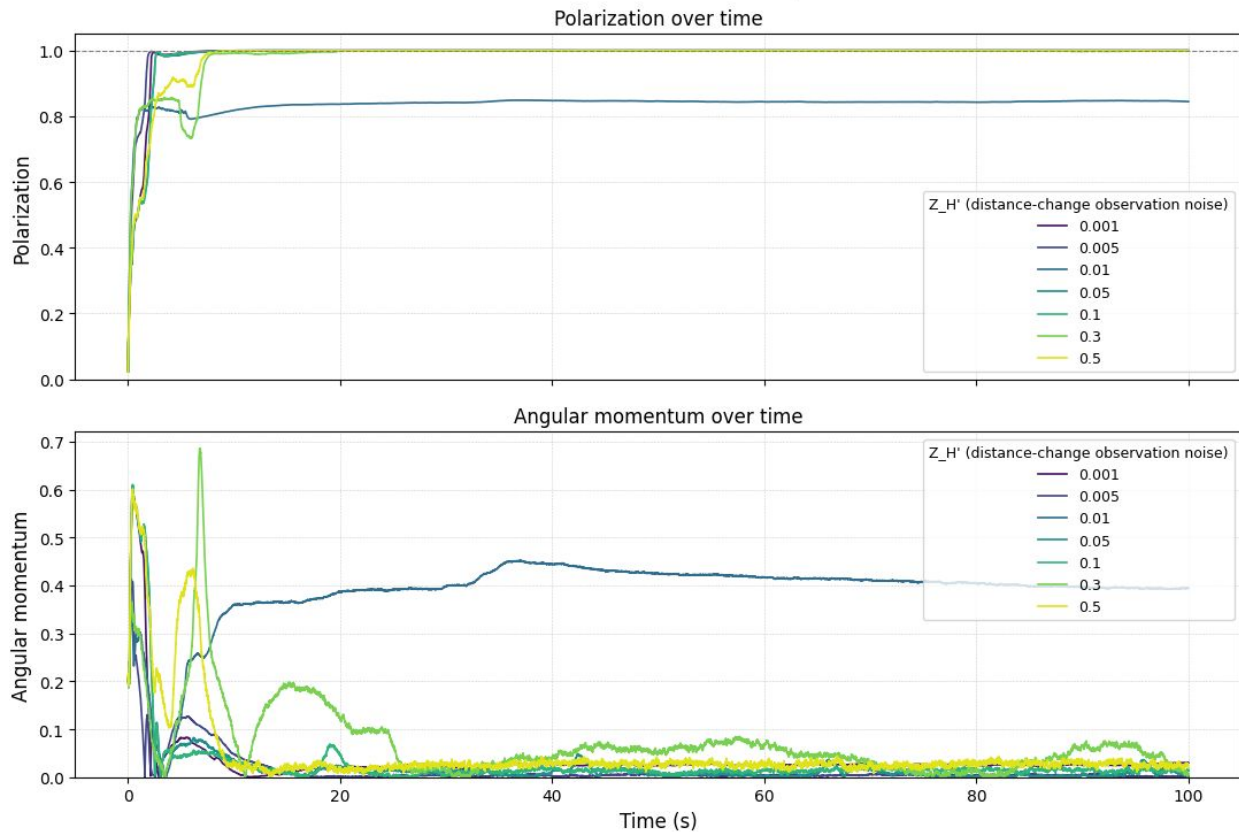
Effect of Z_H (distance-observation noise) on collective behavior
 $T=100s$, initial seed=1, error seed=1; other noise parameters=0.0



Parameter	Mean pol.	Mean ang. mom.
0.001	1.000000	0.006429
0.005	0.999999	0.000587
0.01	1.000000	0.003436
0.05	0.999999	0.007799
0.1	0.999998	0.002701
0.3	0.999992	0.021779
0.5	0.999989	0.022523

Z_H' – $p(t)$ and $m(t)$ plots across time (100 sec) – Avg Neighbor

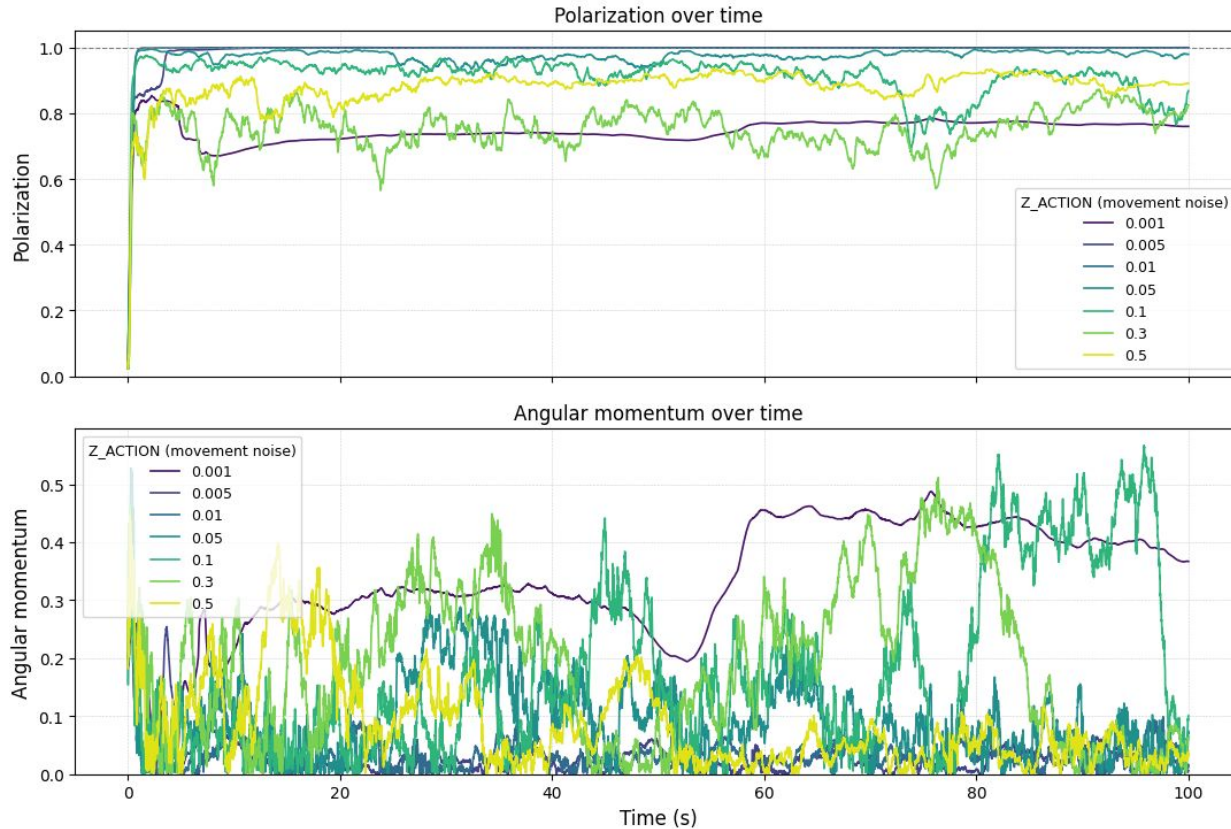
Effect of Z_H' (distance-change observation noise) on collective behavior
T=100s, initial seed=1, error seed=1; other noise parameters=0.0



Parameter	Mean pol.	Mean ang. mom.
0.001	0.999994	0.026628
0.005	0.999993	0.002621
0.01	0.846764	0.394363
0.05	0.999807	0.0097
0.1	0.999947	0.013131
0.3	0.998577	0.05047
0.5	0.99977	0.026663

Z_{Action} – $p(t)$ and $m(t)$ plots across time (100 sec) – Avg Neighbor

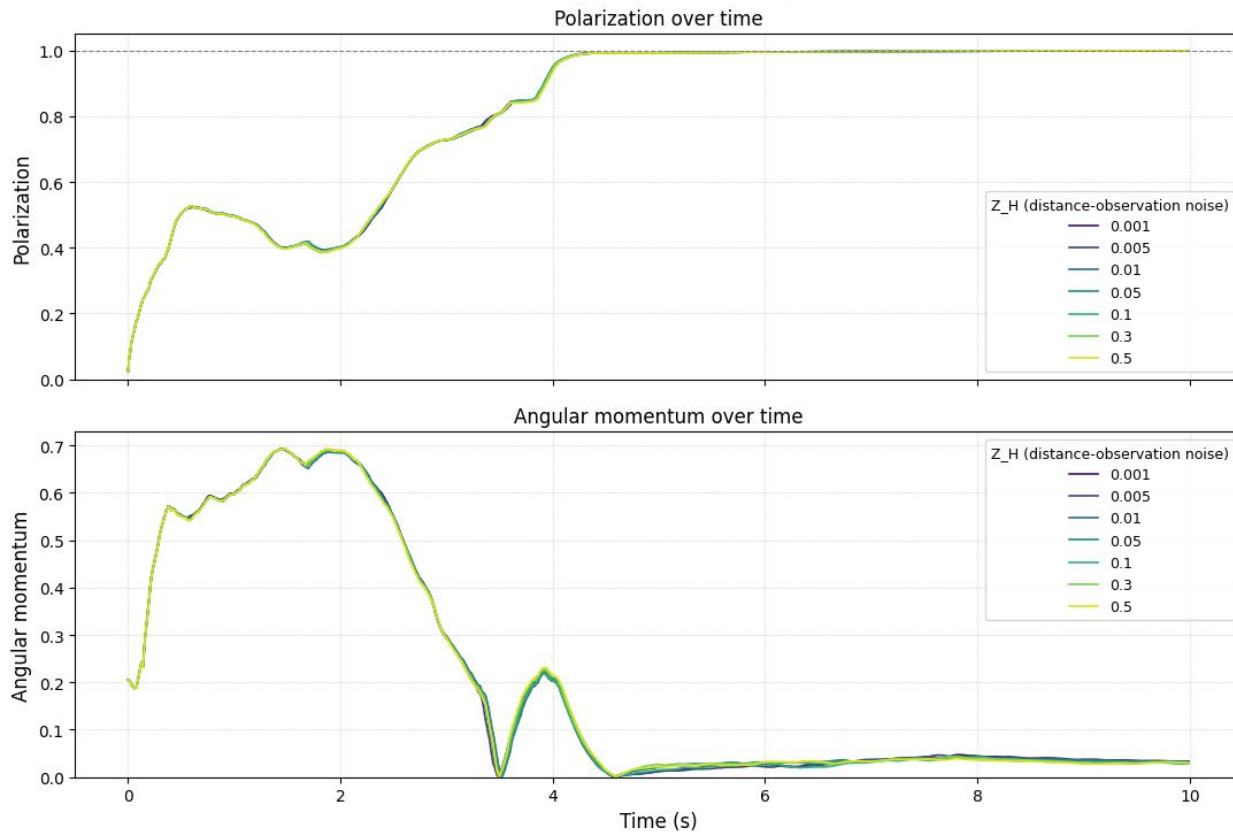
Effect of Z_{ACTION} (movement noise) on collective behavior
 $T=100\text{s}$, initial seed=1, error seed=1; other noise parameters=0.0



Parameter	Mean pol.	Mean ang. mom.
0.001	0.765060	0.392869
0.005	0.999356	0.029416
0.01	0.999007	0.035671
0.05	0.981819	0.039510
0.1	0.865715	0.360465
0.3	0.821567	0.038030
0.5	0.884008	0.038059

$Z_H - p(t)$ and $m(t)$ plots across time (10 sec) – Avg Neighbor

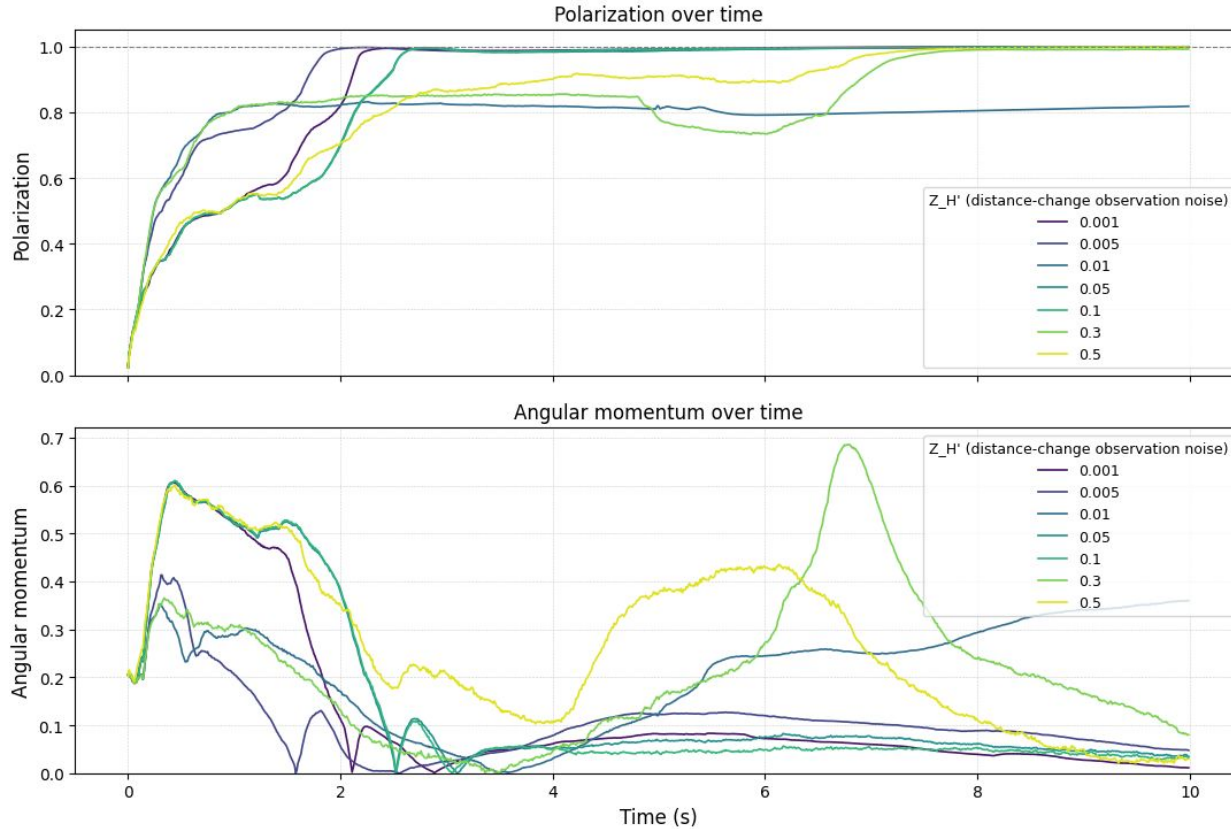
Effect of Z_H (distance-observation noise) on collective behavior
 $T=10s$, initial seed=1, error seed=1; other noise parameters=0.0



Parameter	Mean pol.	Mean ang. mom.
0.001	0.999088	0.034839
0.005	0.999123	0.034487
0.01	0.999085	0.033766
0.05	0.999080	0.033085
0.1	0.999121	0.029942
0.3	0.999166	0.030173
0.5	0.999184	0.029641

Z_H , – $p(t)$ and $m(t)$ plots across time (10 sec) – Avg Neighbor

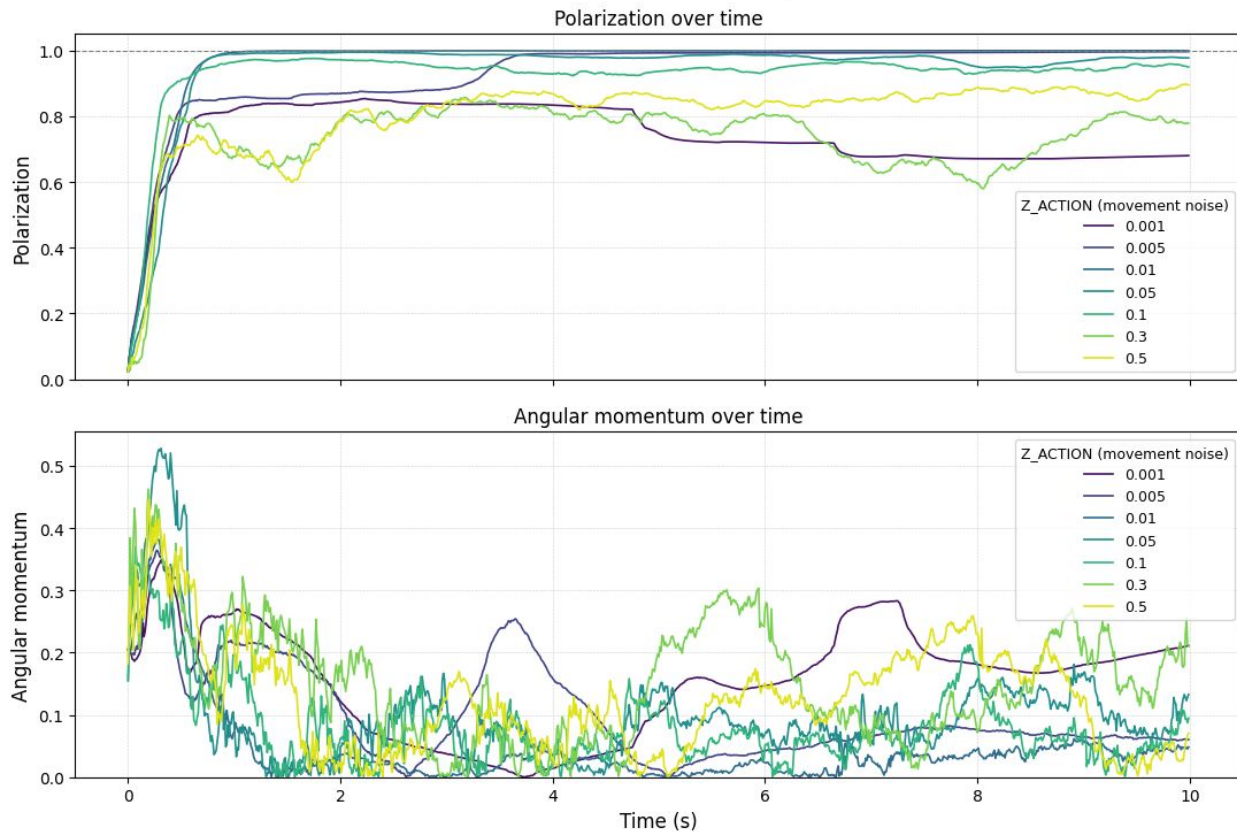
Effect of Z_H ' (distance-change observation noise) on collective behavior
 $T=10s$, initial seed=1, error seed=1; other noise parameters=0.0



Parameter	Mean pol.	Mean ang. mom.
0.001	0.999420	0.018981
0.005	0.998686	0.057800
0.01	0.814871	0.349410
0.05	0.999335	0.041814
0.1	0.999212	0.035001
0.3	0.991112	0.128224
0.5	0.998366	0.028679

Z_{Action} – $p(t)$ and $m(t)$ plots across time (10 sec) – Avg Neighbor

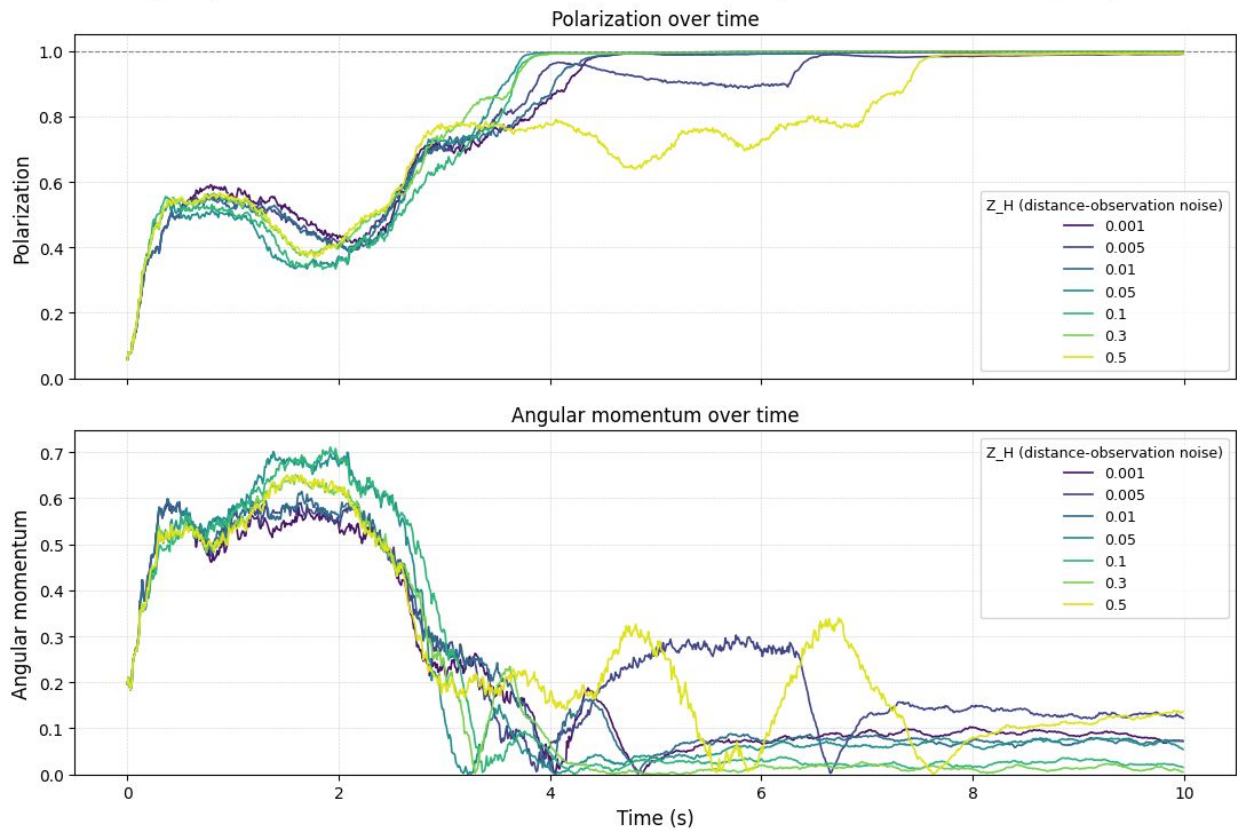
Effect of Z_{ACTION} (movement noise) on collective behavior
 $T=10\text{s}$, initial seed=1, error seed=1; other noise parameters=0.0



Parameter	Mean pol.	Mean ang. mom.
0.001	0.676463	0.195098
0.005	0.996155	0.062820
0.01	0.999426	0.042833
0.05	0.977063	0.097082
0.1	0.950162	0.046900
0.3	0.791655	0.177957
0.5	0.869964	0.038323

Z_H – $p(t)$ and $m(t)$ plots across time (10 sec) – One Neighbor

Effect of Z_H (distance-observation noise) on collective behavior
random neighbor per sector $T=10s$, initial seed=1, error seed=1; other noise parameters=0.0; random neighbor per sector

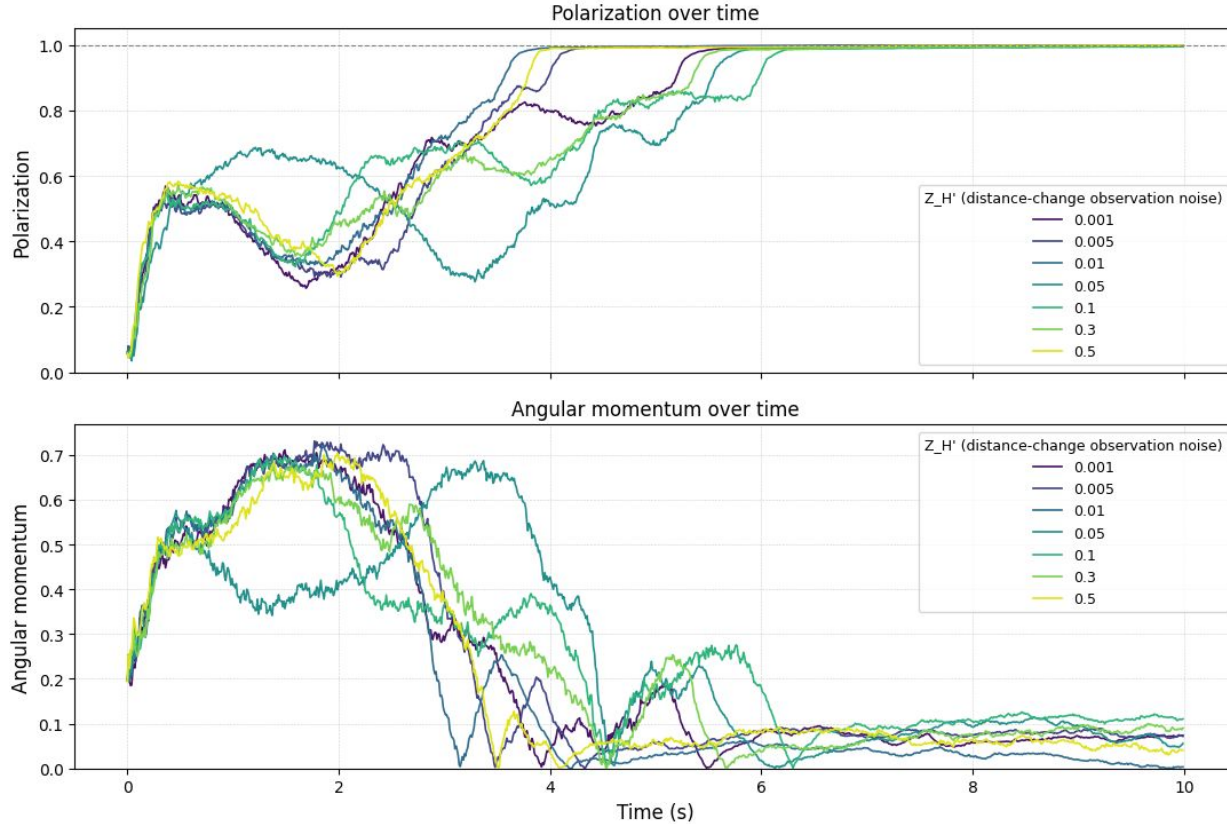


Parameter	Mean pol.	Mean ang. mom.
0.001	0.998229	0.08269
0.005	0.989304	0.128395
0.01	0.998237	0.072996
0.05	0.999177	0.070981
0.1	0.999339	0.024729
0.3	0.999396	0.014123
0.5	0.99176	0.124276

Z_H' – $p(t)$ and $m(t)$ plots across time (10 sec) – Avg Neighbor

Effect of Z_H' (distance-change observation noise) on collective behavior

random neighbor per sector $T=10s$, initial seed=1, error seed=1; other noise parameters=0.0; random neighbor per sector

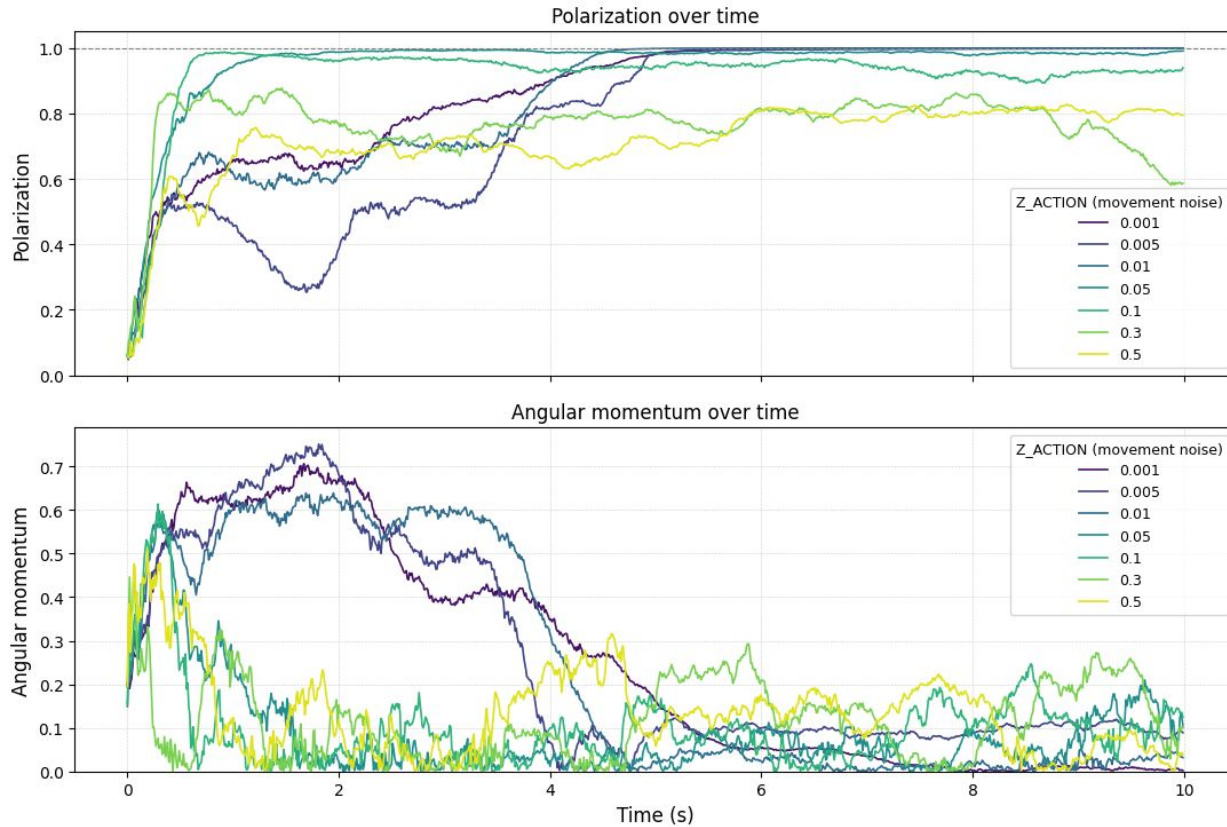


Parameter	Mean pol.	Mean ang. mom.
0.001	0.998058	0.068768
0.005	0.999049	0.076632
0.01	0.999408	0.009286
0.05	0.99711	0.067864
0.1	0.99399	0.111011
0.3	0.996893	0.08826
0.5	0.998942	0.045415

Z_{Action} – $p(t)$ and $m(t)$ plots across time (10 sec) – Avg Neighbor

Effect of Z_{ACTION} (movement noise) on collective behavior

random neighbor per sector $T=10s$, initial seed=1, error seed=1; other noise parameters=0.0; random neighbor per sector



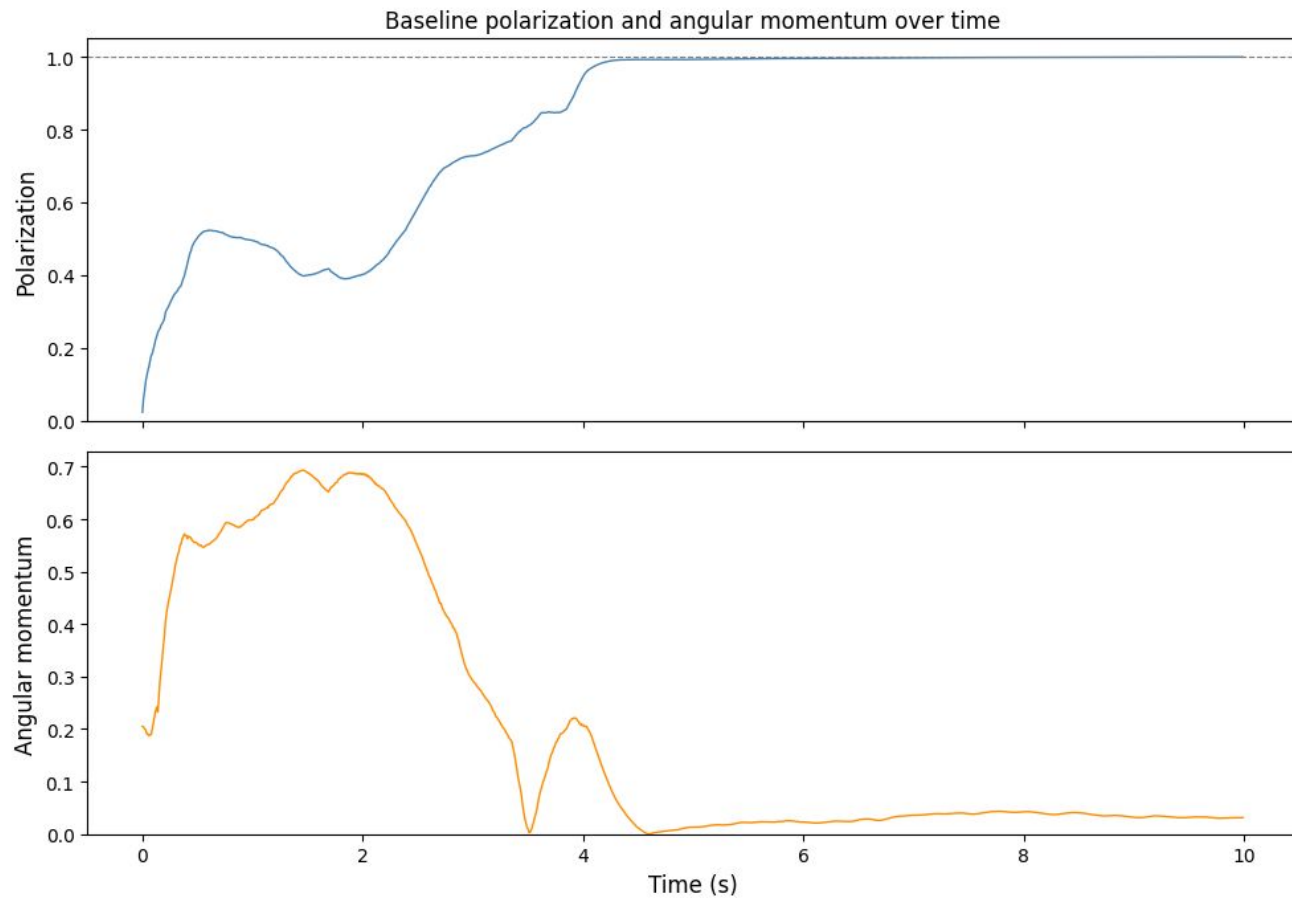
Parameter	Mean pol.	Mean ang. mom.
0.001	0.999541	0.006024
0.005	0.998476	0.099231
0.01	0.999666	0.028843
0.05	0.98371	0.136753
0.1	0.928508	0.098037
0.3	0.685231	0.182726
0.5	0.801409	0.052815

BASELINE $p(t)$ and $m(t)$

SEED_INIT	1	N	30
T	100	DT	0.01
N_SECTORS	4	SECTOR_ANGLE	60.0
DIST_THR	5.0	Z_H	0.0
Z_HPRIME	0.0	Z_ACTION	0.0
ALPHA	0.05	ETA	1.0
PI_Z_SPATIAL	1.0	PI_W_SPATIAL	1.0
S_Z	1.0	S_W	1.0
INFER_LR	0.1	ACTION_LR	0.1
NSTEPS_INFER	1	NSTEPS_ACTION	1
NORMALIZE_V	True	SEED_ERROR	1

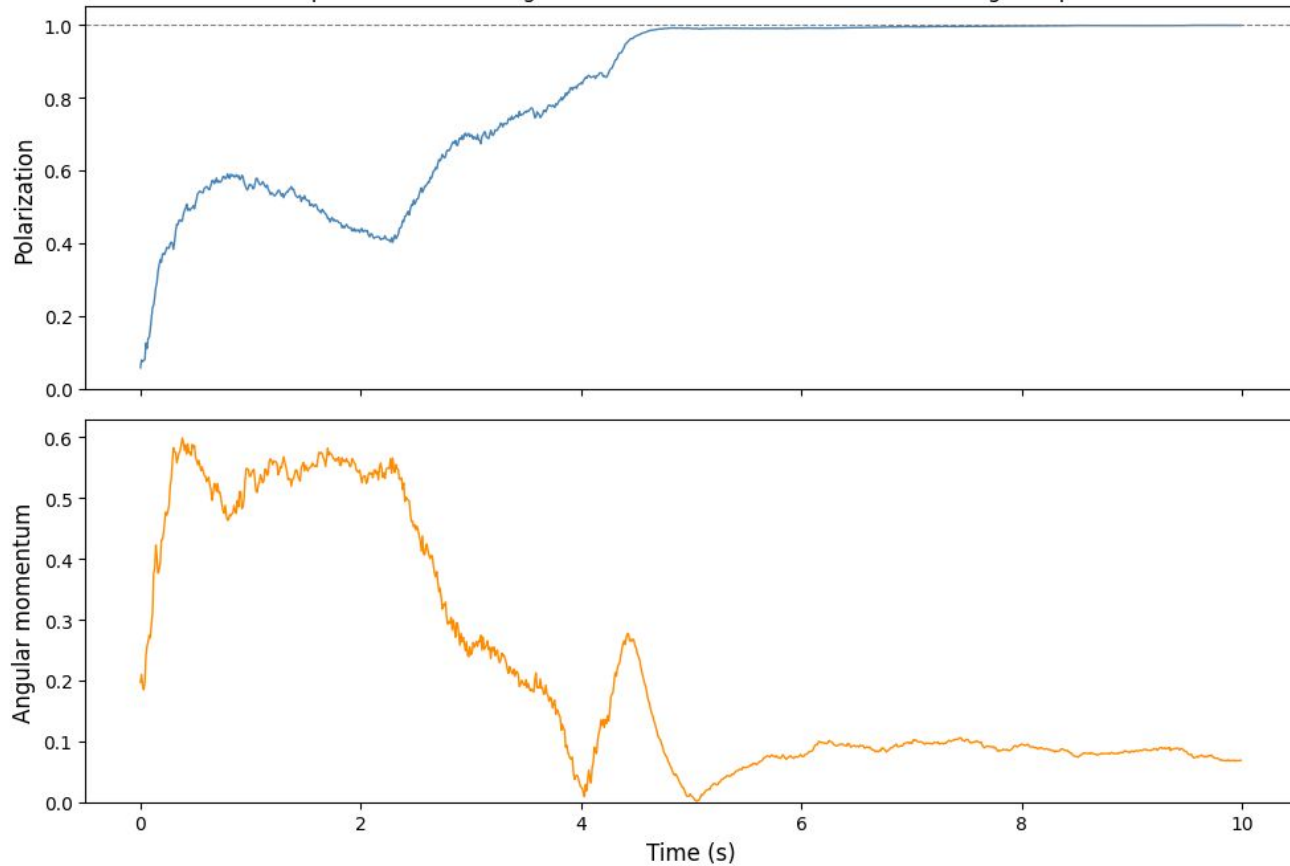
The left table contains the parameter set used to consider the baseline runs for the average neighbor and one neighbor runs.

BASELINE $p(t)$ and $m(t)$ – Avg Neighbor



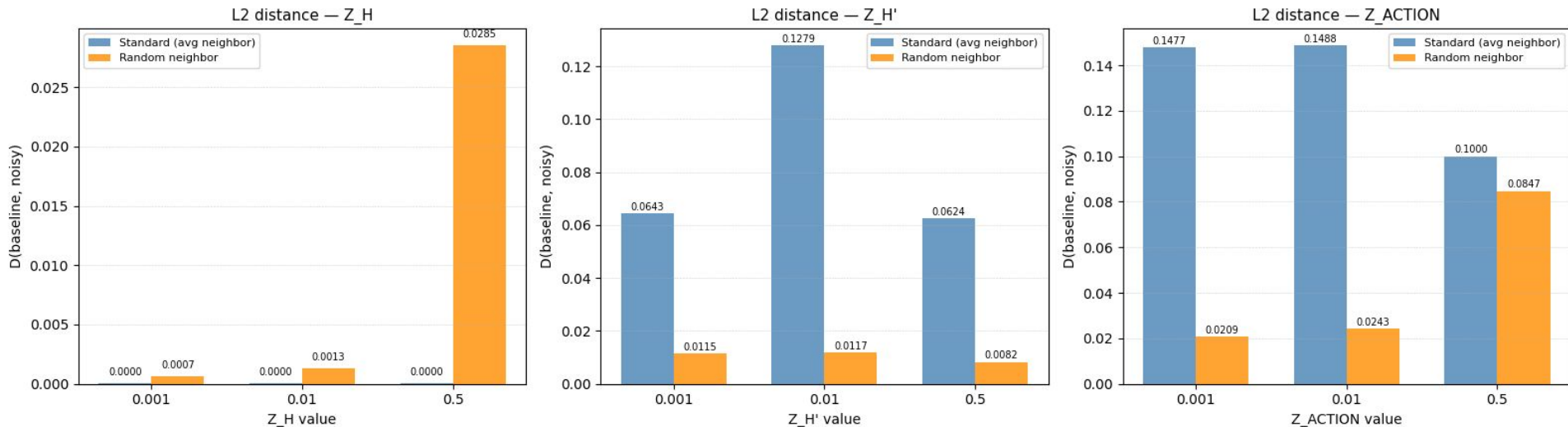
BASLINE $p(t)$ and $m(t)$ – One Neighbor

Baseline polarization and angular momentum over time -- Random neighbor per sector



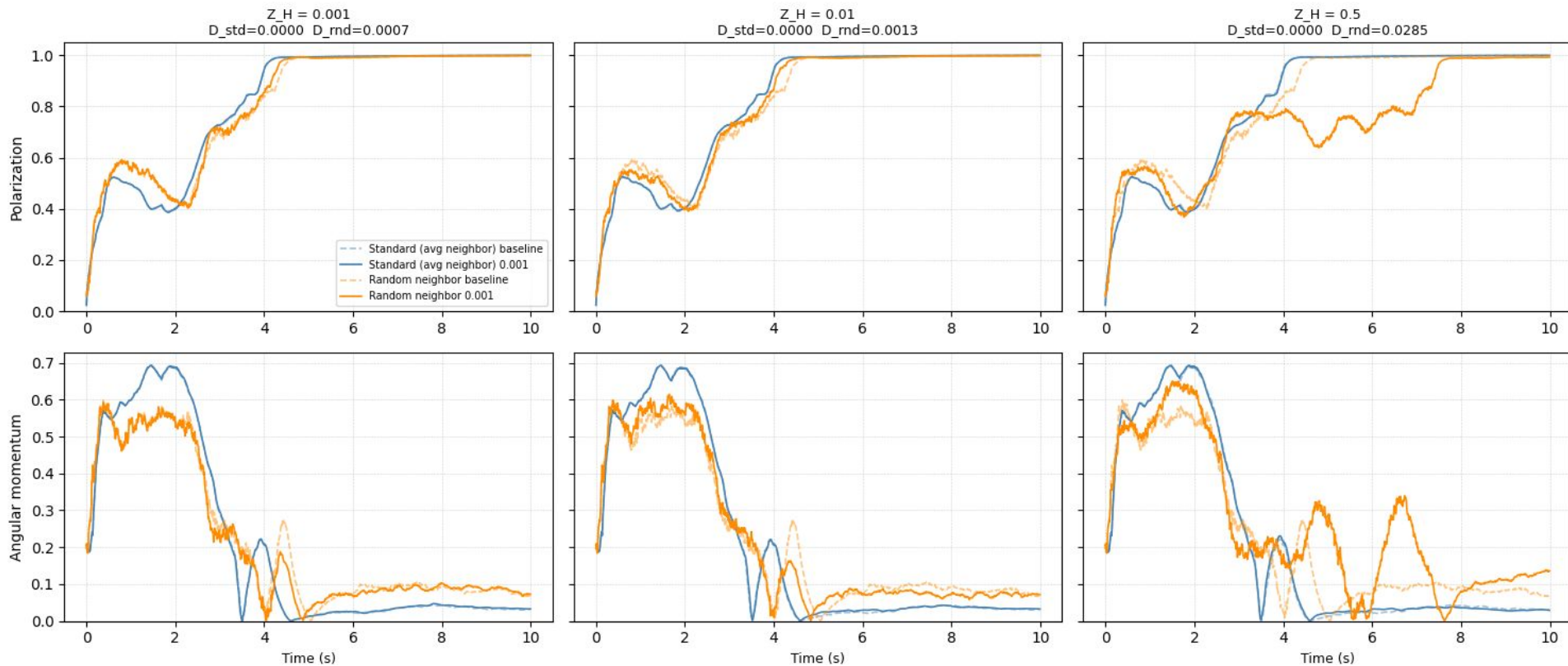
Comparison between baseline and Z noise values

Trajectory divergence from zero-noise baseline
$$D(i,j) = (1/T) \int ||f_{\text{baseline}}(t) - f_{\text{noisy}}(t)||^2 dt \quad \text{where } f(t) = [\text{polarization}(t), \text{angular_momentum}(t)]$$



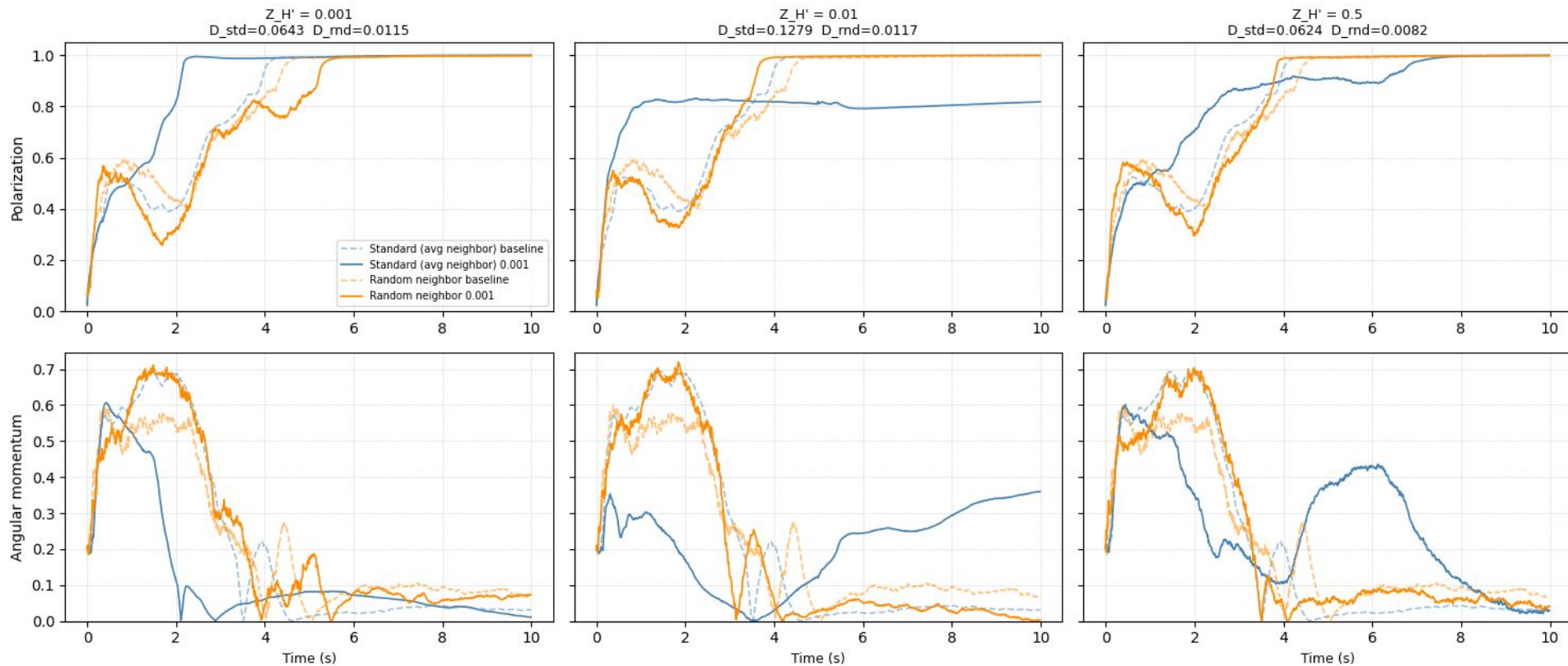
Comparison between baseline and Z noise values

Baseline vs noisy trajectories — Z_H
Dashed = zero-noise baseline, Solid = noisy run | $D = (1/T) \int ||f_{\text{base}} - f_{\text{noisy}}||^2 dt$



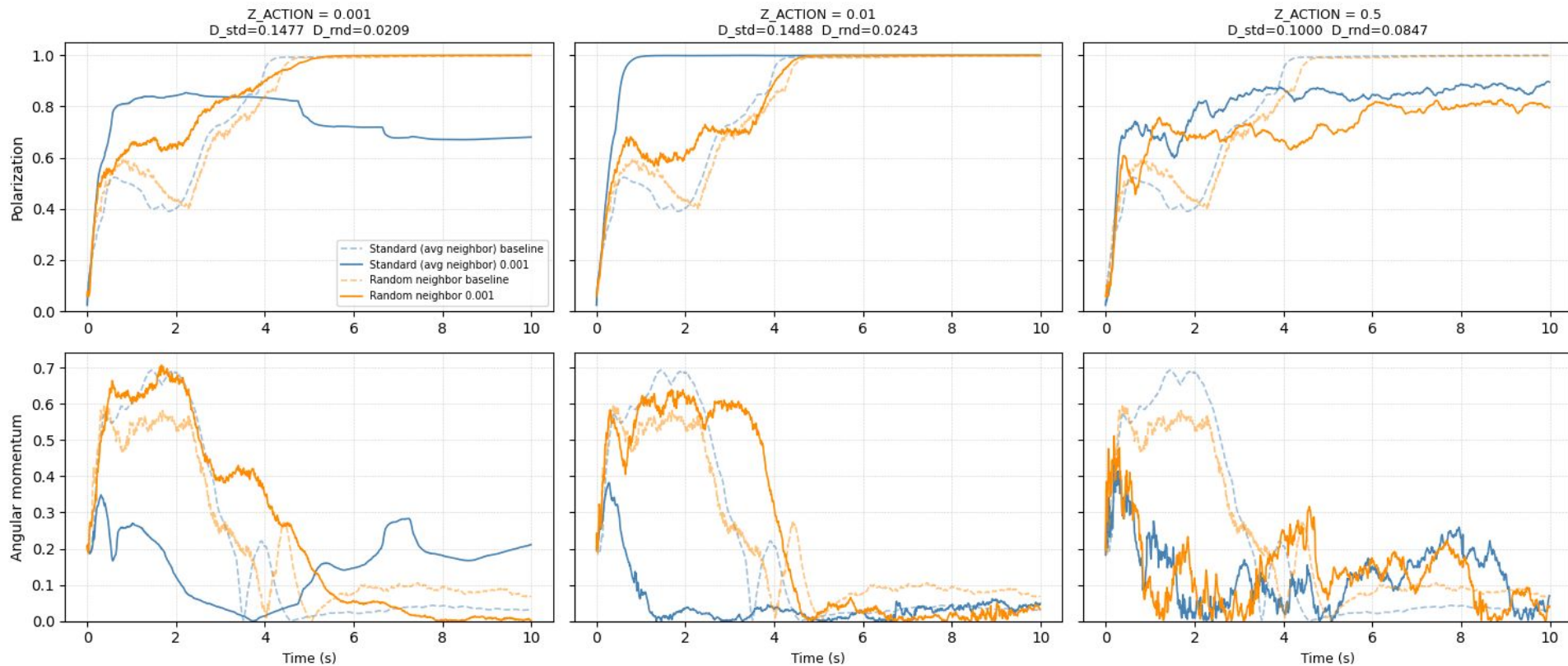
Comparison between baseline and Z noise values

Baseline vs noisy trajectories — $Z_{H'}$
Dashed = zero-noise baseline, Solid = noisy run | $D = (1/T) \int ||f_{\text{base}} - f_{\text{noisy}}||^2 dt$



Comparison between baseline and Z noise values

Baseline vs noisy trajectories — Z_ACTION
Dashed = zero-noise baseline, Solid = noisy run | $D = (1/T) \int ||f_{\text{base}} - f_{\text{noisy}}||^2 dt$



Comparison between baseline and Z noise values

L2 distance from zero-noise baseline				
Noise type	Value	D (standard	D (random	Ratio (rnd/std)
Z_H	0.001	0.000011	0.000666	61.2448
Z_H	0.01	0.000021	0.001291	60.917
Z_H	0.5	0.000046	0.028539	615.6245
Z_H'	0.001	0.064331	0.011536	0.1793
Z_H'	0.01	0.127929	0.011745	0.0918
Z_H'	0.5	0.06244	0.008203	0.1314
Z_ACTION	0.001	0.147746	0.020867	0.1412
Z_ACTION	0.01	0.148791	0.024293	0.1633
Z_ACTION	0.5	0.100009	0.08474	0.8473